Image-Identifiable Genomic Subspaces: A Linear–Gaussian Model of Radiogenomic Recoverability

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Abstract

How much of a patient's genomic state can be read off their scan? We answer this with a number, in bits, that a study of a given size cannot exceed. Starting from a linear–Gaussian channel in which the genomic state $g \in \mathbb{R}^p$ is Gaussian and the imaging phenotype $x \in \mathbb{R}^d$ is a noisy linear readout of it, the classical theory gives the *channel* information $I(g;x)$ and identifies the maximally recoverable genomic directions with canonical correlation analysis. That is an infinite-data quantity, and it is not what a cohort can demonstrate. Our main contribution is the finite-sample counterpart: for any learning algorithm trained on n patients with a $d^\star$-dimensional imaging representation, the attainable out-of-sample recoverability of a genomic direction obeys $\mathcal{R}_n = \rho^2\, n/(n+\nu)$ with $\nu = d^\star(1-\rho^2)/\rho^2$, giving an *attainable information* $\mathcal{I}_n = -\frac{1}{2}\sum_i \log(1-\mathcal{R}_{n,i})$ that is zero at $n = 0$ and reaches $I(g;x)$ only as $n \to \infty$. Two consequences are immediate: the learning cost ν is simply the feature count divided by the channel signal-to-noise ratio, and attainable signal scales as ρ^4 rather than ρ^2, so weak genomic directions are doubly penalized and cannot be certified at realistic n. We verify the bound in simulation — six model families, from ridge to gradient boosting to a neural network, approach it and none crosses it — and then report it for three cohorts using a model-free upper confidence limit obtained by planting signal of known strength into permuted real data and sweeping the planted rank. Bounding a single scalar spectral functional, the fourth moment $\sum_i \rho_i^4$, and converting it by a concavity argument that is exact over every spectrum, the limit applies to *total* retention across all directions rather than to the single best axis, and without any assumption on the shape of the spectrum. At their actual sizes, TCGA-KIRC ($n = 190$, CT), an NSCLC cohort ($n = 129$, CT), and ADNI ($n = 702$, MRI) admit at most 0.76, 0.18 and 1.20 bits per patient about the transcriptome — around one perfectly measured binary biomarker apiece, with the NSCLC statistic in fact falling below its own permutation null — and equivalently a ceiling of $\mathrm{AUC} \leq 0.68$–0.75 on *any* classifier for *any* binarized genomic feature, which is where the published per-gene AUCs in this literature in fact sit. Stress-testing the bound against every combination of training-set size, genomic feature set and learning algorithm — 360 configurations across the three cohorts — no combination crosses it, while 59 of 60 remain non-zero, so the ceiling is neither violated nor vacuous. Reaching 90% of even these channels would take $n \approx 10^3$. Finally, a chain-rule decomposition against hand-specified anchor genes shows the recoverable subspace is not spread across the transcriptome: in both tumor cohorts a four-gene panel of lineage and hypoxia markers accounts for most of the attainable information. The practical message for radiogenomics is that the useful target is a handful of genes and a reportable bit-count, not a transcriptome-wide search.

Keywords radiogenomics · canonical correlation analysis · mutual information · recoverability · linear–Gaussian model

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1 Introduction

Radiogenomics asks how much of a molecular state is written into its image [1–4]. Clinically, useful applications include those where knowledge of a molecular state would change management, including some cancers and neurodegenerative diseases, but the cost or risk of sequencing is prohibitive. For instance, in clear cell renal cell carcinoma, CT descriptors of margin, enhancement and intratumoral vascularity have been tied to *VHL* and to the chromatin-modifying genes that co-occur with it [5–8]. In diffuse glioma, MR appearance carries *IDH1* status and 1p/19q codeletion well enough to support classifiers built on imaging alone [9, 10]. In lung adenocarcinoma, CT phenotypes have been used to call *EGFR* mutation status [11–13].

None of these mappings is a lookup table from a single imaging sign to a single variant. Whatever molecular state reaches the image reaches it diffusely, through texture, shape, and spatial heterogeneity, so the field moved quickly from hand-specified radiomic descriptors [14–16] to representations learned from the images themselves [17, 18]. Convolutional networks with residual connections [19] are trained end to end on volumes, and one such network classifies *IDH1*, 1p/19q and MGMT status directly from MRI [20], as another does for *EGFR* from CT [13]. Encoder–decoder segmentation networks supply the region of interest and increasingly the features as well [21, 22]; vision transformers [23] model context across a whole volume rather than a local receptive field; attention-based multiple-instance pooling aggregates slices or patches into one patient-level prediction [24]; and large pretrained encoders are now used as general-purpose imaging feature extractors [25, 26].

What almost none of this work asks is how much there was to find. The standard output is a classifier and an AUC, and the usual reading is asymmetric: a weak result is blamed on the model, a strong one is credited to biology. At the cohort sizes typical of the field ($n \sim 10^2$) against panels of 10^3–10^4 transcripts, that reading is unsafe in both directions. This is a high-dimensional, low-power multiple-testing problem in which apparent positives are easily manufactured by overfitting, batch structure, or demographic confounding [27, 28]. Absent a ceiling there is no way to say whether an AUC of 0.7 is near the limit of what the scan encodes or a tenth of it, whether the next gain should come from more patients or a better architecture, or when a search has been exhausted. We argue that the more answerable question is not "does imaging predict gene j?" but *how many genomic directions are image-identifiable at all, and how well* — a low-dimensional, estimable quantity with a hard ceiling set by the biology–imaging channel rather than by the size of the gene panel.

This paper supplies that ceiling. We adopt the simplest model with a non-trivial answer: a linear–Gaussian channel from genomics $g \in \mathbb{R}^p$ to imaging $x \in \mathbb{R}^d$. Under it every quantity of interest is closed-form: the Bayes-optimal estimator $\mathbb{E}[g \mid x]$, the posterior covariance, the mutual information $I(g; x)$, and a per-direction *recoverability score* $R(v) \in [0, 1]$. The directions of maximal recoverability coincide exactly with canonical correlation analysis (CCA) between the two modalities, with $R(a_i) = \rho_i^2$, the squared canonical correlation, so a vague "fundamental limit" intuition becomes a concrete estimable spectrum $\rho_1^2 \geq \rho_2^2 \geq \cdots$ whose directions can be read back onto genes and pathways. That spectrum is an infinite-data object, however, and it is not what a cohort can demonstrate: the map from imaging to genomics must itself be learned from the same patients, and learning it costs information. Our central contribution is to price that cost exactly. For *any* learning algorithm trained on n patients with a $d^\star$-dimensional imaging representation, the attainable out-of-sample recoverability of a genomic direction cannot exceed $\mathcal{R}_n = \rho^2 n/(n + \nu)$ with learning cost $\nu = d^\star(1 - \rho^2)/\rho^2$, so the extractable information is capped by an *attainable information* $\mathcal{I}_n$ that is zero with no data and reaches the channel value $I(g; x)$ only in the limit. The distinction between $I(g; x)$ and $\mathcal{I}_n$ is the practical content of the paper: the first is what radiogenomics is about, the second is what a study of a given size can ever show, and at $n \sim 10^2$ they differ by an order of magnitude. Two consequences reframe how the field should spend its effort. The learning cost ν is the imaging feature count divided by the channel's signal-to-noise ratio, so the sample size required scales with the representation one chooses, and there is an optimal, finite feature budget (Corollary 4). And because $\mathcal{R}_n$ decays like ρ^4 rather than ρ^2 for weak channels (Corollary 2), marginal genomic directions are doubly penalized: widening a gene panel buys precisely the directions a cohort of that size cannot certify.

The paper is organized as follows. Sections 3–11 develop the channel, the recoverability score and its CCA eigenproblem, and the rank limit on how many directions are identifiable at all. Section 12 proves the finite-n ceiling and checks in simulation that six model families approach it and none crosses it. Section 13 treats estimation from finite samples, including the sample-size regimes in which the spectrum collapses and the permutation and cross-validation controls we use throughout, and Section 15 extends the framework to binary mutation calls via a Gaussian copula, which converts bits into a per-gene AUC ceiling. Section 16 sets up a chain-rule decomposition against hand-specified anchor genes: where a real channel exists, its information should concentrate in a handful of canonical genes rather than spreading across the transcriptome. Sections 17–19 apply all of this to TCGA-KIRC, an NSCLC cohort, and ADNI, and Section 20 reports the ceiling for the three at their actual sizes and stress-tests it against 360 combinations of training-set size, genomic feature set and learning algorithm. The anchor-gene concentration appears where there is a real channel to concentrate, and, informatively, fails to appear in the cohorts whose apparent signal is overfit or confounded, which makes it a diagnostic as well as a summary.

2 Related work

CCA and its probabilistic form. Canonical correlation analysis dates to Hotelling [29]; its probabilistic / latent-variable interpretation, which we use in Section 13.4, is due to Bach and Jordan [30], and the kernel/regularized variants are surveyed by Hardoon et al. [31]. The identity that links Gaussian mutual information to the canonical correlations, $I = -\frac{1}{2}\sum_i \log(1-\rho_i^2)$, is classical [32]. Our recoverability score and eigenproblem are a re-reading of this machinery in terms of posterior-variance reduction per genomic direction; we add the rank limit and the binary-data extension.

High-dimensional and sparse CCA. When p, d are comparable to n, sample canonical correlations are severely biased upward. The limiting laws of sample canonical correlations under the null and finite-rank alternatives are characterized by random-matrix theory [33–35], which we use directly to place a detection edge on the spectrum (Section 13.5). Two families of remedies exist: regularization/shrinkage (Ledoit–Wolf [36]) and sparsity (penalized CCA, 37; sparse-PCA consistency theory, 38). We adopt symmetric dimension reduction plus shrinkage and lean on cross-validation and permutation for honesty rather than on sparsity, but sparse CCA is a natural alternative estimator within the same framing.

Information-theoretic limits and estimation. The fundamental-limit perspective — bounding what *any* predictor can achieve — is in the spirit of rate–distortion theory [39] and the information bottleneck [40–42], specialized here to a linear–Gaussian channel where the bottleneck is exactly solvable. Because we report mutual information, we are mindful that general MI estimation is fragile in high dimensions and that variational estimators can be badly biased [43–45]; our reported MI is the closed-form Gaussian functional of estimated canonical correlations, not a generic estimator, and we bound its misspecification error in both directions (Sections 13.7, 17.4).

Radiogenomics. Empirical radiogenomics has reported imaging correlates of molecular state across many tumor types [5–8, 11, 17, 18], typically as per-gene or per-signature predictors built on radiomic features [14–16] or deep embeddings [25]. Our framework is complementary: it does not build a predictor but bounds and counts the directions any predictor could exploit, and it supplies the confound and sample-size controls that distinguish a real low-rank channel from the overfitting and batch effects that the per-gene paradigm is prone to.

3 Setup and notation

We observe n patients. For patient i let $g_i \in \mathbb{R}^p$ be a vector of genomic features (e.g. expression, somatic mutation burden, copy-number scores) and $x_i \in \mathbb{R}^d$ a vector of imaging phenotypes (radiomic descriptors, deep features, or structured reads) [2, 14, 46]. Stack the observations into data matrices $\mathbf{G} \in \mathbb{R}^{n\times p}$ and $\mathbf{X} \in \mathbb{R}^{n\times d}$, with row i equal to $g_i^\top$ and $x_i^\top$ respectively. Random variables are written in uppercase (G, X), realizations in lowercase (g, x). Throughout we assume both modalities have been centered, so all means are zero; recentring is handled by the estimators of Section 13.

We use $\Sigma_G \in \mathbb{R}^{p\times p}$, $\Sigma_X \in \mathbb{R}^{d\times d}$ for the marginal covariances, $\Sigma_{GX} = \mathrm{Cov}(G, X) \in \mathbb{R}^{p\times d}$ for the cross-covariance, and $\Sigma_{XG} = \Sigma_{GX}^\top$. For a symmetric positive semidefinite S we write $S^{1/2}$ for its symmetric square root and $S^{-1/2}$ for the inverse thereof.

4 The linear–Gaussian generative model

The genomic state is drawn from a Gaussian prior and the imaging phenotype is a noisy linear readout of it:

$$G \sim \mathcal{N}(0, \Sigma_G), \tag{1}$$

$$X \mid G \sim \mathcal{N}(AG,\ \sigma^2 I_d), \qquad A \in \mathbb{R}^{d\times p}. \tag{2}$$

The matrix A encodes how genomic variation is expressed in the image; the isotropic term $\sigma^2 I_d$ collects everything in the image that genomics does not drive — biological variation outside the measured panel, acquisition and reconstruction noise, and feature-extraction error. Equivalently,

$$X = AG + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d), \quad \varepsilon \perp G. \tag{3}$$

Equations (1)–(3) are deliberately the simplest model with a non-trivial answer. Their value is that *every* downstream quantity is closed-form, so the model functions as an exact upper bound: no estimator — linear or nonlinear — can recover genomics from imaging better than the posterior derived below (under these assumptions). This fundamental-limit framing is in the spirit of rate–distortion theory and the information bottleneck [39–42, 47], specialized here to a tractable linear–Gaussian channel. Section 13 relaxes the isotropic-noise and known-parameter assumptions for practical fitting.

5 Marginal and joint laws

Proposition 1 (Joint Gaussian law; classical Gaussian identity, e.g. 32)**.** *Under* (1)–(3)*,*

$$\begin{pmatrix} G \\ X \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \Sigma_G & \Sigma_G A^\top \\ A\Sigma_G & A\Sigma_G A^\top + \sigma^2 I_d \end{pmatrix}\right). \tag{4}$$

In particular,

$$X \sim \mathcal{N}(0, \Sigma_X), \qquad \Sigma_X = A\Sigma_G A^\top + \sigma^2 I_d, \tag{5}$$

and the cross-covariance is

$$\Sigma_{GX} = \mathrm{Cov}(G, X) = \Sigma_G A^\top. \tag{6}$$

Proof. Since G and ε are independent Gaussians and (G, X) is the affine image $(G,\ AG + \varepsilon)$, the pair is jointly Gaussian with zero mean. The covariance blocks follow by standard Gaussian conditioning: $\mathrm{Cov}(G, G) = \Sigma_G$, $\mathrm{Cov}(G, X) = \mathrm{Cov}(G, AG + \varepsilon) = \Sigma_G A^\top$ (using $\varepsilon \perp G$), and $\mathrm{Var}(X) = \mathrm{Cov}(AG + \varepsilon) = A\Sigma_G A^\top + \sigma^2 I_d$. □

Equation (5) is the model's first interpretive statement: imaging covariance decomposes additively into a *genomics-driven* component $A\Sigma_G A^\top$ and an *irreducible* component $\sigma^2 I_d$. The cross-covariance $\Sigma_{GX} = \Sigma_G A^\top$ is the central estimable object — it names which genomic directions co-vary with imaging at all.

6 Bayes-optimal recovery of genomics from imaging

Proposition 2 (Posterior; standard Gaussian conditioning)**.** *The posterior of genomics given imaging is Gaussian,* $G \mid X \sim \mathcal{N}(\hat{g}(X), \Sigma_{G|X})$*, with*

$$\hat{g}(X) = \mathbb{E}[G \mid X] = \Sigma_G A^\top \big(A\Sigma_G A^\top + \sigma^2 I_d\big)^{-1} X \;=\; BX, \tag{7}$$

$$\Sigma_{G|X} = \Sigma_G - \Sigma_G A^\top \big(A\Sigma_G A^\top + \sigma^2 I_d\big)^{-1} A\Sigma_G. \tag{8}$$

Equivalently, in precision form,

$$\Sigma_{G|X}^{-1} = \Sigma_G^{-1} + \sigma^{-2} A^\top A, \qquad B = \sigma^{-2}\, \Sigma_{G|X}\, A^\top. \tag{9}$$

Proof. By standard Gaussian conditioning on the joint law of Proposition 1, $G \mid X$ is Gaussian with mean $\Sigma_{GX}\Sigma_X^{-1}X$ and covariance $\Sigma_G - \Sigma_{GX}\Sigma_X^{-1}\Sigma_{XG}$. Substituting the blocks $\Sigma_{GX} = \Sigma_G A^\top$, $\Sigma_{XG} = A\Sigma_G$, $\Sigma_X = A\Sigma_G A^\top + \sigma^2 I_d$ gives (7)–(8) and $B = \Sigma_G A^\top \Sigma_X^{-1}$. The precision form $\Sigma_{G|X}^{-1} = \Sigma_G^{-1} + \sigma^{-2}A^\top A$ and $B = \sigma^{-2}\Sigma_{G|X}A^\top$ follow from the Woodbury identity applied to $\Sigma_{G|X}$ and the push-through identity $A^\top(A\Sigma_G A^\top + \sigma^2 I_d) = \sigma^2(\Sigma_G^{-1} + \sigma^{-2}A^\top A)\Sigma_G A^\top$. □

The estimator $\hat{g}(X) = BX$ is the Bayes-optimal predictor under squared-error loss; because the model is jointly Gaussian it is also the optimal predictor over *all* measurable functions of X, not merely linear ones. Thus, *under the assumed linear–Gaussian law*, $\Sigma_{G|X}$ is a hard floor on residual genomic uncertainty: **no algorithm can do better**. This is a statement about the model, not an unconditional guarantee on real data; Section 13.7 quantifies how far each cohort departs from the law and what that costs the bound. The precision form (9) makes the accounting transparent — imaging *adds* information $\sigma^{-2}A^\top A$ to the prior precision Σ_G^{-1}.

7 Mutual information and the SNR decomposition

Proposition 3 (Mutual information; classical Gaussian MI identity, 32)**.** *The mutual information between the two modalities is*

$$I(G;X) = \tfrac{1}{2}\log\frac{\det\Sigma_G}{\det\Sigma_{G|X}} = \tfrac{1}{2}\log\det\big(I_d + \sigma^{-2}A\Sigma_G A^\top\big). \tag{10}$$

Let $\lambda_1 \geq \cdots \geq \lambda_r > 0$ be the nonzero eigenvalues of $A\Sigma_G A^\top$. Then

$$I(G;X) = \tfrac{1}{2}\sum_{i=1}^{r}\log\left(1+\frac{\lambda_i}{\sigma^2}\right). \tag{11}$$

Proof. By standard Gaussian conditioning the mutual information of a jointly Gaussian pair is $I(G;X) = \frac{1}{2}\log(\det\Sigma_G/\det\Sigma_{G|X})$ [32]. Taking determinants in the precision form (9) and factoring out Σ_G^{-1} gives $\det\Sigma_G/\det\Sigma_{G|X} = \det(I_p + \sigma^{-2}\Sigma_G^{1/2}A^\top A\,\Sigma_G^{1/2})$; Sylvester's identity $\det(I_p+PQ) = \det(I_d+QP)$ converts this to $\det(I_d + \sigma^{-2}A\Sigma_G A^\top)$, which is (10). Diagonalizing the symmetric PSD matrix $A\Sigma_G A^\top$ with nonzero eigenvalues λ_i, the determinant is $\prod_i(1+\lambda_i/\sigma^2)$, giving (11). □

Equation (11) is the cleanest reading of the model: each image-visible genomic direction contributes $\frac{1}{2}\log(1+\mathrm{SNR}_i)$ bits (nats), where $\mathrm{SNR}_i = \lambda_i/\sigma^2$ is the signal-to-noise ratio of that direction. Directions whose genomic signal is small relative to imaging noise contribute essentially nothing, regardless of how many genes load on them. Figure 1 illustrates the decomposition (11) on the synthetic instance introduced below: the per-direction bit contribution falls off rapidly after the first handful of canonical components, and the cumulative total saturates at $I(G;X) \approx 6.45$ bits out of the analytic upper bound implied by the model.

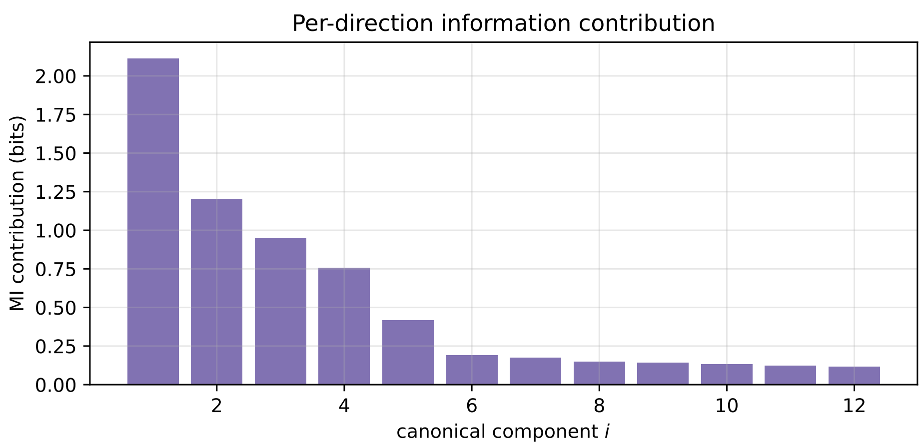


Figure 1: Per-direction contribution to $I(G;X)$ on a synthetic linear–Gaussian instance ($n = 800$, $p = 80$, $d = 40$). Each bar is $\frac{1}{2}\log(1+d_i^2)$ in bits; the dashed line is the cumulative total.

8 Canonical coordinates and the recoverability spectrum

The eigenvalues λ_i are most interpretable after whitening. Define the whitened genomic variable $\tilde{G} = \Sigma_G^{-1/2}G \sim \mathcal{N}(0, I_p)$ and the *whitened transfer operator*

$$M := \sigma^{-1}A\,\Sigma_G^{1/2} \in \mathbb{R}^{d\times p}, \qquad M = \sum_i d_i\,u_i v_i^\top \quad \text{(SVD)}, \tag{12}$$

with left/right singular vectors $u_i \in \mathbb{R}^d$, $v_i \in \mathbb{R}^p$ and singular values $d_1 \geq d_2 \geq \cdots \geq 0$.

Proposition 4 (Canonical decomposition of recoverability; recombination of Hotelling CCA 29 and probabilistic CCA 30)**.** *With M as in* (12)*, define the genomic canonical direction $a_i := \Sigma_G^{-1/2}v_i$ and the canonical score $S_i := a_i^\top G = v_i^\top \tilde{G}$. Then:*

(i) S_i *has prior variance* 1 *and posterior variance* $\mathrm{Var}(S_i \mid X) = (1 + d_i^2)^{-1}$*;*

(ii) the canonical correlation between S_i *and imaging is* $\rho_i = d_i/\sqrt{1 + d_i^2}$*, equivalently* $d_i^2 = \rho_i^2/(1 - \rho_i^2)$*;*

(iii) the eigenvalues of Section 7 satisfy $\lambda_i/\sigma^2 = d_i^2$*, so*

$$I(G; X) = \tfrac{1}{2} \sum_i \log(1 + d_i^2) = -\tfrac{1}{2} \sum_i \log(1 - \rho_i^2). \tag{13}$$

Proof. Write the SVD as $M = UDV^\top$ with $U^\top U = I$, $V^\top V = I$, and $D = \mathrm{diag}(d_1, d_2, \dots)$, so $V = [v_1, v_2, \dots]$ collects the right singular vectors and u_i the left ones. Since $S_i = v_i^\top \tilde{G}$ with $\tilde{G} \sim \mathcal{N}(0, I_p)$, the prior variance is $\mathrm{Var}(S_i) = v_i^\top v_i = 1$.

Posterior variance (i). In whitened coordinates the mean readout is $AG = A\Sigma_G^{1/2}\tilde{G} = \sigma M\tilde{G}$, so the likelihood is $X \mid \tilde{G} \sim \mathcal{N}(\sigma M\tilde{G},\ \sigma^2 I_d)$. Applying the precision form (9) to the pair $(\tilde{G}, X)$ — whose prior covariance is I_p and whose transfer matrix is σM with noise $\sigma^2 I_d$ — the posterior precision of $\tilde{G}$ is

$$\mathrm{Var}(\tilde{G} \mid X)^{-1} = I_p + \sigma^{-2}(\sigma M)^\top(\sigma M) = I_p + M^\top M = V\,(I_p + D^2)\,V^\top,$$

using $M^\top M = VD^2V^\top$. Inverting and applying $V^\top V = I_p$,

$$\mathrm{Var}(\tilde{G} \mid X) = V\,(I_p + D^2)^{-1}V^\top,$$

and therefore

$$\mathrm{Var}(S_i \mid X) = v_i^\top\,\mathrm{Var}(\tilde{G} \mid X)\,v_i = \big[(I_p + D^2)^{-1}\big]_{ii} = \frac{1}{1 + d_i^2},$$

since the v_i are orthonormal. This proves (i).

Canonical correlation (ii). The cross-covariance of the whitened genomics with imaging is

$$\mathrm{Cov}(\tilde{G}, X) = \mathbb{E}\big[\tilde{G}\,(\sigma M\tilde{G} + \varepsilon)^\top\big] = \sigma\,\mathbb{E}[\tilde{G}\tilde{G}^\top]\,M^\top + \mathbb{E}[\tilde{G}\varepsilon^\top] = \sigma M^\top,$$

using $\mathbb{E}[\tilde{G}\tilde{G}^\top] = I_p$ and $\tilde{G} \perp \varepsilon$. Hence $\mathrm{Cov}(S_i, X) = v_i^\top\,\mathrm{Cov}(\tilde{G}, X) = \sigma v_i^\top M^\top = \sigma d_i u_i^\top$, the last step because $v_i^\top M^\top = (Mv_i)^\top = (d_i u_i)^\top$ from the SVD. Pair S_i with the matched imaging variate $T_i := u_i^\top X$. From $\Sigma_X = \sigma^2(MM^\top + I_d)$ (Proposition 1 in whitened form) and $MM^\top u_i = d_i^2 u_i$,

$$\mathrm{Var}(T_i) = u_i^\top \Sigma_X u_i = \sigma^2\big(u_i^\top MM^\top u_i + 1\big) = \sigma^2(1 + d_i^2),$$

while $\mathrm{Cov}(S_i, T_i) = \mathrm{Cov}(S_i, X)\,u_i = \sigma d_i u_i^\top u_i = \sigma d_i$. The correlation is therefore

$$\rho_i = \mathrm{corr}(S_i, T_i) = \frac{\mathrm{Cov}(S_i, T_i)}{\sqrt{\mathrm{Var}(S_i)\ \mathrm{Var}(T_i)}} = \frac{\sigma d_i}{\sqrt{1}\,\sqrt{\sigma^2(1 + d_i^2)}} = \frac{d_i}{\sqrt{1 + d_i^2}},$$

and rearranging gives $d_i^2 = \rho_i^2/(1 - \rho_i^2)$, proving (ii).

Connection to the eigenvalues (iii). From $M = \sigma^{-1}A\Sigma_G^{1/2}$ we get $A\Sigma_G A^\top = \sigma^2 MM^\top$, whose nonzero eigenvalues are $\sigma^2 d_i^2$; comparing with the λ_i of Proposition 3 gives $\lambda_i = \sigma^2 d_i^2$, i.e. $\lambda_i/\sigma^2 = d_i^2$. Substituting into (11) and using $1 + d_i^2 = 1/(1 - \rho_i^2)$ from (ii),

$$I(G; X) = \tfrac{1}{2} \sum_i \log(1 + d_i^2) = -\tfrac{1}{2} \sum_i \log(1 - \rho_i^2),$$

which is (iii). □

Part (i) gives the headline identity. Define the recoverability of a genomic score as one minus the fraction of its variance that survives observing imaging (formalized in Section 9); then

$$\boxed{R(a_i) = 1 - \mathrm{Var}(S_i \mid X) = 1 - \frac{1}{1 + d_i^2} = \frac{d_i^2}{1 + d_i^2} = \rho_i^2.} \tag{14}$$

The recoverability of the i-th canonical genomic direction is exactly the squared canonical correlation. This is what reduces the whole analysis to CCA between $\mathbf{G}$ and $\mathbf{X}$ (Section 13), and it means A and σ^2 never have to be estimated individually — only the three second moments $\Sigma_G, \Sigma_X, \Sigma_{GX}$ enter.

9 Recoverability of a named genomic direction

In practice one often cares about a specific, biologically named genomic score $Y = v^\top G$ for a fixed $v \in \mathbb{R}^p$ — a pathway signature, a polygenic score, a single gene's expression ($v = e_j$).

Definition 1 (Recoverability score)**.** *For $v \in \mathbb{R}^p$ with $v^\top \Sigma_G v > 0$,*

$$R(v) := 1 - \frac{v^\top \Sigma_{G|X}\, v}{v^\top \Sigma_G\, v} = \frac{v^\top (\Sigma_G - \Sigma_{G|X})\, v}{v^\top \Sigma_G\, v} \;\in\; [0,1]. \tag{15}$$

$R(v) = 0$ means imaging carries no information about $Y = v^\top G$; $R(v) = 1$ means imaging determines it exactly. The score is scale-invariant in v and, by the Gaussian conditional-variance identity, equals the population R^2 of the Bayes-optimal predictor of Y from X: $R(v) = 1 - \mathbb{E}[\operatorname{Var}(Y \mid X)]/\operatorname{Var}(Y)$. The matrix at the core of (15) has the convenient moment-only form

$$\Sigma_G - \Sigma_{G|X} = \Sigma_G A^\top \Sigma_X^{-1} A \Sigma_G = \Sigma_{GX}\, \Sigma_X^{-1}\, \Sigma_{XG}. \tag{16}$$

10 The recoverability eigenproblem

The natural project-level question — *which* genomic directions does imaging see best — is the variational problem

$$\max_{v \neq 0}\; R(v) = \max_{v \neq 0}\; \frac{v^\top \big(\Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}\big) v}{v^\top \Sigma_G\, v}. \tag{17}$$

Proposition 5 (Recoverability eigenproblem = CCA; the generalized eigenproblem is Hotelling's 29)**.** *The stationary points of* (17) *are the solutions of the generalized eigenproblem*

$$\Sigma_{GX}\, \Sigma_X^{-1}\, \Sigma_{XG}\; v = \mu\, \Sigma_G\, v. \tag{18}$$

Its eigenvalues are $\mu_i = \rho_i^2 = d_i^2/(1+d_i^2)$ and its eigenvectors are the canonical genomic directions a_i of Proposition 4, Σ_G-orthonormal ($a_i^\top \Sigma_G a_j = \delta_{ij}$). Thus the ordered solutions of (18) *are precisely the canonical correlation directions between G and X, and $\max_v R(v) = \rho_1^2$.*

Proof. Write $C := \Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}$ for the (symmetric positive-semidefinite) numerator matrix, so the objective is the generalized Rayleigh quotient $R(v) = v^\top C v / v^\top \Sigma_G v$. Setting its gradient to zero,

$$\nabla_v R(v) = \frac{2Cv\,(v^\top \Sigma_G v) - 2\Sigma_G v\,(v^\top C v)}{(v^\top \Sigma_G v)^2} = 0 \quad\iff\quad Cv = \frac{v^\top C v}{v^\top \Sigma_G v}\, \Sigma_G v = \mu\, \Sigma_G v,$$

with $\mu = R(v)$ the value of the quotient at the stationary point. This is the generalized eigenproblem (18).

To solve it, whiten via $v = \Sigma_G^{-1/2} w$. Left-multiplying (18) by $\Sigma_G^{-1/2}$ and inserting $\Sigma_G^{-1/2}\Sigma_G^{1/2} = I$ turns it into the ordinary symmetric eigenproblem

$$\big(\Sigma_G^{-1/2} C\, \Sigma_G^{-1/2}\big)\, w = \mu\, w.$$

Now expand C. Using $\Sigma_{GX} = \Sigma_G A^\top$ and $\Sigma_{XG} = A\Sigma_G$,

$$\Sigma_G^{-1/2} C\, \Sigma_G^{-1/2} = \Sigma_G^{-1/2}\,(\Sigma_G A^\top)\, \Sigma_X^{-1}\,(A\Sigma_G)\, \Sigma_G^{-1/2} = \big(\Sigma_G^{1/2} A^\top\big)\, \Sigma_X^{-1}\, \big(A \Sigma_G^{1/2}\big).$$

Substituting $A\Sigma_G^{1/2} = \sigma M$ and $\Sigma_X = \sigma^2 (MM^\top + I_d)$,

$$\Sigma_G^{1/2} A^\top\, \Sigma_X^{-1}\, A \Sigma_G^{1/2} = (\sigma M^\top) \cdot \sigma^{-2} (MM^\top + I_d)^{-1} \cdot (\sigma M) = M^\top (MM^\top + I_d)^{-1} M.$$

Insert the SVD $M = UDV^\top$. Then $MM^\top + I_d = UD^2U^\top + I_d$, and on the range of U its inverse acts as $U(D^2+I)^{-1}U^\top$, so

$$M^\top (MM^\top + I_d)^{-1} M = VDU^\top \cdot U(D^2+I)^{-1}U^\top \cdot UDV^\top = V\, D^2 (D^2+I)^{-1}\, V^\top,$$

using $U^\top U = I$. This is an eigendecomposition: the eigenvalues are $d_i^2/(1+d_i^2)$ with eigenvectors $w = v_i$. By Proposition 4(ii) these eigenvalues equal ρ_i^2, so $\mu_i = \rho_i^2$ and the leading value is $\max_v R(v) = \mu_1 = \rho_1^2$.

Undoing the whitening, the genomic eigenvector is $v = \Sigma_G^{-1/2} w = \Sigma_G^{-1/2} v_i = a_i$, the canonical direction of Proposition 4. Finally, the Σ_G-orthonormality follows because the v_i are orthonormal:

$$a_i^\top \Sigma_G\, a_j = (\Sigma_G^{-1/2} v_i)^\top \Sigma_G\, (\Sigma_G^{-1/2} v_j) = v_i^\top v_j = \delta_{ij}.$$

□

So the deliverable of the analysis is a *recoverability spectrum* $\rho_1^2 \geq \rho_2^2 \geq \cdots$ together with the genomic directions a_i that achieve it. Reading the loadings of $a_1, a_2, \ldots$ back onto genes or pathways answers "which biology is image-identifiable": the leading directions typically align with coarse, spatially expressed programs — proliferation, hypoxia, angiogenesis, necrosis, immune infiltration, tumor burden, stromal remodeling — rather than with individual transcripts, consistent with clear-cell RCC radiogenomic findings [5, 7, 48].

Illustration on synthetic data. Figure 2 shows the recoverability spectrum on a synthetic radiogenomic instance generated from the linear–Gaussian model of Section 4 ($n = 800$, $p = 80$, $d = 40$, latent rank $k = 6$ with prescribed $\{\rho_i^\star\}$, the last of which is near zero). The estimated $\hat{\rho}_i^2$ closely match the planted values for the identifiable directions and decay toward the high-dimensional bulk thereafter, validating Proposition 5. Figure 3 reads the same estimator against the analytic ground truth derived from Propositions 2–4: the recovered posterior mean $\hat{B}X$, posterior-variance retention, and direction alignment all track the closed-form predictions.

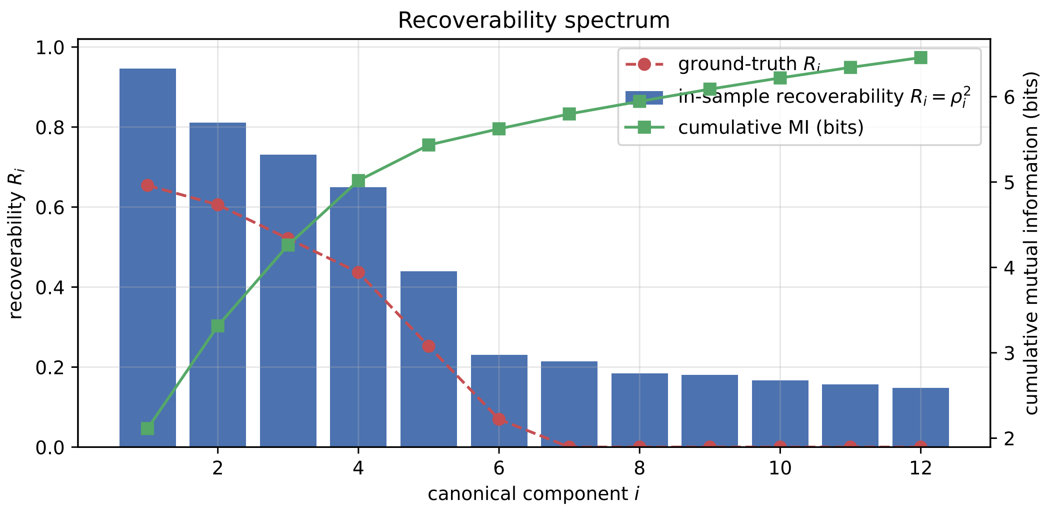


Figure 2: Recoverability spectrum on a synthetic linear–Gaussian instance ($n = 800$, $p = 80$, $d = 40$). The leading $\hat{\rho}_i^2$ recovers the planted canonical structure; later components fall to the high-dimensional bulk.

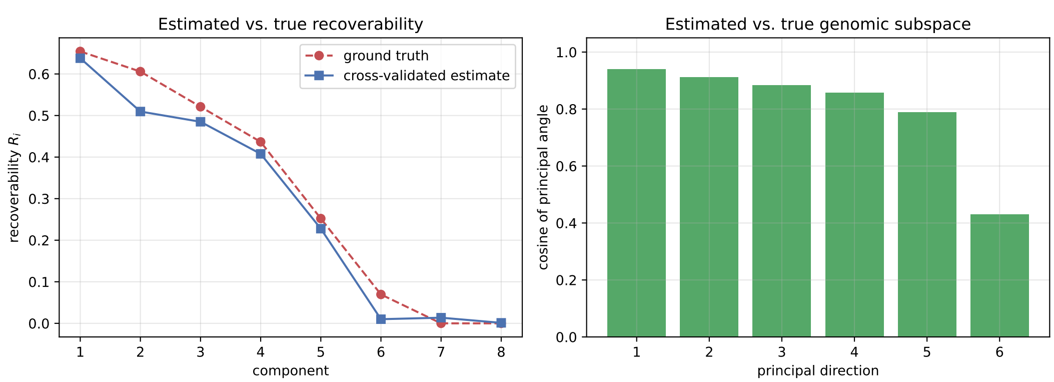


Figure 3: Empirical-vs-analytic check on the same synthetic instance. Estimated canonical correlations, recoverability per direction, and imaging variance attributable to genomics all match the closed-form quantities from Propositions 2–4.

11 Rank limits and identifiability

Proposition 6 (Rank limit)**.** *The recoverable signal matrix has rank*

$$\operatorname{rank}(\Sigma_G - \Sigma_{G|X}) = \operatorname{rank}(A\Sigma_G A^\top) \leq \min\big(d,\ p,\ \operatorname{rank} A,\ \operatorname{rank} \Sigma_G\big). \tag{19}$$

Consequently at most $\min(d, p, \operatorname{rank} A, \operatorname{rank} \Sigma_G)$ *canonical correlations are nonzero, and the image-identifiable genomic subspace* $\operatorname{span}\{a_i : \rho_i > 0\}$ *has dimension bounded by the same quantity.*

This is the conceptual core. Even with $p = 20{,}000$ measured genes, imaging can expose only a projection of genomics whose dimension is capped by the rank of the transfer map A — if a CT phenotype reflects, say, 20 macroscopic biological axes, then no model recovers 20,000 independent molecular variables from it. The recoverable dimension is a property of the *biology–imaging channel*, not of the estimator. A data-processing remark: imaging features are themselves a deterministic, lossy function $x = \phi(\text{image})$; any such post-processing can only shrink $\operatorname{rank}(\Sigma_{GX})$ and lower each ρ_i, never raise them, by the data processing inequality [32, 42, 49–51].

12 The attainable-information ceiling

Proposition 6 bounds what the *channel* permits: with infinite data, imaging exposes a genomic subspace of dimension at most $\min(d, p, \operatorname{rank} A, \operatorname{rank} \Sigma_G)$, and the information carried is $I(G; X) = -\frac{1}{2}\sum_i \log(1 - \rho_i^2)$. That is a statement about biology and imaging physics. It is not the number a radiogenomic study can report, because no study has infinite data: the map from imaging to genomics must itself be *learned* from the same cohort, and learning it costs information.

This section supplies the missing ceiling. We ask: given n patients and a $d^\star$-dimensional imaging representation, what is the largest out-of-sample recoverability — and hence the largest mutual information — that *any* learning algorithm can achieve? The answer is a closed-form function of $(\rho^2, n, d^\star)$ alone, it vanishes at $n = 0$, and it rises to the channel value ρ^2 only as $n \to \infty$. It is the object we propose the field should report.

12.1 The learning problem

Fix a genomic target $Y = v^\top G$ with $\operatorname{Var}(Y) = 1$ (for a canonical direction $v = a_i$ this normalization is automatic, Proposition 4(i)). Under the joint Gaussian law of Proposition 1,

$$Y = \beta^\top X + \eta, \qquad \beta = \Sigma_X^{-1}\Sigma_{XY} \in \mathbb{R}^{d^\star}, \qquad \eta \perp X, \quad \operatorname{Var}(\eta) = 1 - \rho^2, \tag{20}$$

where $\rho^2 = R(v)$ is the channel recoverability of the target. A *learning algorithm* $\mathcal{A}$ maps a training sample $D_n = \{(g_j, x_j)\}_{j=1}^n$ to a predictor $\hat f_{D_n} : \mathbb{R}^{d^\star} \to \mathbb{R}$; it may be linear or arbitrarily nonlinear.

Definition 2 (Attainable recoverability)**.** *The recoverability* attained *by $\mathcal{A}$ at sample size n is*

$$R_n(\mathcal{A}) := 1 - \frac{\mathbb{E}_{D_n}\, \mathbb{E}_{(Y,X)}\big[\big(Y - \hat f_{D_n}(X)\big)^2\big]}{\operatorname{Var}(Y)}, \tag{21}$$

the expectation over a fresh patient (Y, X) independent of D_n. The inner expectation is the out-of-sample R^2 of the fitted predictor; the outer one averages over training sets.

Definition 2 is exactly what a cross-validated R^2 estimates, which is what makes the bound below directly comparable to reported numbers rather than to an idealization.

To bound R_n over *all* algorithms we must say what the learner does not know. The linear–Gaussian model constrains the total signal, $\beta^\top\Sigma_X\beta = \rho^2$, but says nothing about *which* imaging directions carry it: the model is invariant under Σ_X-orthogonal rotations of the imaging coordinates. The unique prior with that invariance and the correct second moment is Zellner's g-prior with g matched to the channel signal-to-noise ratio,

$$\beta \;\sim\; \mathcal{N}\Big(0,\; \frac{\rho^2}{d^\star}\,\Sigma_X^{-1}\Big), \qquad \text{so that} \qquad \mathbb{E}\big[\operatorname{Var}(\beta^\top X)\big] = \frac{\rho^2}{d^\star}\operatorname{tr}(\Sigma_X^{-1}\Sigma_X) = \rho^2. \tag{22}$$

12.2 The ceiling

Proposition 7 (Attainable-recoverability ceiling)**.** *Under* (20)–(22)*, every learning algorithm $\mathcal{A}$ satisfies, on average over the signal direction,*

$$\boxed{\mathbb{E}_\beta\, R_n(\mathcal{A}) \;\le\; \mathcal{R}_n(\rho^2) \;:=\; \rho^2\,\frac{n}{n+\nu}, \qquad \nu \;:=\; \frac{d^\star\,(1-\rho^2)}{\rho^2} \;=\; \frac{d^\star}{\mathrm{SNR}},} \tag{23}$$

with $\mathrm{SNR} = \rho^2/(1-\rho^2) = d_i^2$ as in Proposition 4. The bound holds even if the learner is told ρ^2, Σ_X and $\operatorname{Var}(\eta)$ exactly. It is attained, with equality, by ridge regression with penalty ν in whitened imaging coordinates whenever the design is balanced ($\mathbf{X}^\top\mathbf{X} = n\Sigma_X$).

Proof. Write $\operatorname{risk}(\mathcal{A}, \beta)$ for the inner expectation in (21), so that $\mathbb{E}_\beta R_n(\mathcal{A}) = 1 - \mathbb{E}_\beta \operatorname{risk}(\mathcal{A}, \beta)$ under $\operatorname{Var}(Y) = 1$. For any fixed prior, the prior-averaged risk of every algorithm is at least the Bayes risk, $\mathbb{E}_\beta \operatorname{risk}(\mathcal{A}, \beta) \ge \operatorname{risk}^{\text{Bayes}}$, so a lower bound on the Bayes risk gives the claimed upper bound on $\mathbb{E}_\beta R_n$ uniformly in $\mathcal{A}$. It therefore suffices to bound the Bayes risk below. Since the model is jointly Gaussian, the Bayes predictor of Y given a fresh x and the training set is $\mathbb{E}[Y \mid x, D_n] = \bar\beta(D_n)^\top x$, where $\bar\beta$ is the posterior mean of β. Conditioning on X and using $\eta \perp X$,

$$\mathbb{E}\big[(Y - \bar\beta^\top X)^2\big] = \mathrm{Var}(\eta) + \mathbb{E}\big[(\bar\beta - \beta)^\top \Sigma_X (\bar\beta - \beta)\big] = (1-\rho^2) + \mathbb{E}\ \mathrm{tr}\big(\Sigma_X\, \Sigma_{\mathrm{post}}\big),$$

because for the Gaussian linear model the posterior covariance

$$\Sigma_{\mathrm{post}} = \Big(\tfrac{1}{1-\rho^2}\,\mathbf{X}^\top\mathbf{X} + \tfrac{d^\star}{\rho^2}\,\Sigma_X\Big)^{-1}$$

depends on the design $\mathbf{X}$ but not on the responses. The map $S \mapsto S^{-1}$ is operator convex on positive-definite matrices, so by Jensen's inequality $\mathbb{E}[\Sigma_{\mathrm{post}}] \succeq \big(\frac{n}{1-\rho^2}\Sigma_X + \frac{d^\star}{\rho^2}\Sigma_X\big)^{-1}$ (using $\mathbb{E}[\mathbf{X}^\top\mathbf{X}] = n\Sigma_X$), and taking the trace against the positive-semidefinite Σ_X preserves the inequality:

$$\mathbb{E}\ \mathrm{tr}(\Sigma_X\Sigma_{\mathrm{post}})\ \ge\ \mathrm{tr}\bigg(\Sigma_X\cdot\Big(\tfrac{n}{1-\rho^2} + \tfrac{d^\star}{\rho^2}\Big)^{-1}\Sigma_X^{-1}\bigg) = \frac{d^\star}{\frac{n}{1-\rho^2} + \frac{d^\star}{\rho^2}} = \frac{d^\star\rho^2(1-\rho^2)}{n\rho^2 + d^\star(1-\rho^2)}.$$

Substituting into (21) with $\mathrm{Var}(Y) = 1$,

$$\mathbb{E}_\beta R_n(\mathcal{A})\ \le\ 1 - (1-\rho^2) - \frac{d^\star\rho^2(1-\rho^2)}{n\rho^2 + d^\star(1-\rho^2)}\ =\ \rho^2\cdot\frac{n\rho^2}{n\rho^2 + d^\star(1-\rho^2)}\ =\ \rho^2\,\frac{n}{n+\nu},$$

which is (23). When $\mathbf{X}^\top\mathbf{X} = n\Sigma_X$ the Jensen step is an equality and the posterior mean is exactly the ridge estimator $\big(\mathbf{X}^\top\mathbf{X} + \nu\,\Sigma_X\big)^{-1}\mathbf{X}^\top\mathbf{y}$, i.e. ridge with penalty ν after whitening, so the bound is attained. □

Reading the ceiling. Equation (23) is a hyperbola in n with three properties worth stating plainly.

- $\mathcal{R}_0 = 0$ and $\mathcal{R}_n \uparrow \rho^2$: with no data nothing is recoverable, and the channel value is approached but never reached.
- ν is the *learning cost* of the direction — the cohort size at which half the channel recoverability is attainable — and it is simply the number of imaging features divided by the channel's signal-to-noise ratio. Weak channels are expensive in exact proportion to how weak they are.
- ν is simultaneously the optimal ridge penalty. The regularization a practitioner tunes by cross-validation is, under this model, an estimate of the same quantity that sets the sample-size requirement.

Solving $\mathcal{R}_n = (1-\delta)\rho^2$ gives the design formula

$$n_{1-\delta}\ =\ \frac{1-\delta}{\delta}\cdot\frac{d^\star(1-\rho^2)}{\rho^2}, \tag{24}$$

so reaching 90% of a channel with $\rho^2 = 0.1$ using $d^\star = 20$ imaging features requires $n \approx 1600$ patients — an order of magnitude above the cohorts in Sections 17–19.

Corollary 1 (Attainable information)**.** *Because the canonical residuals $S_i - \rho_i T_i$ are uncorrelated across i with variances $1-\rho_i^2$ (proof of Proposition 4), the per-direction problems decouple and the ceilings add. No predictor trained on n patients conveys more than*

$$\boxed{\mathcal{I}_n\ =\ -\tfrac{1}{2}\sum_{i=1}^{k}\log\Big(1 - \rho_i^2\,\tfrac{n}{n+\nu_i}\Big), \qquad \nu_i = \frac{d^\star(1-\rho_i^2)}{\rho_i^2},} \tag{25}$$

about the genomic state of a held-out patient, with $\mathcal{I}_0 = 0$ and $\mathcal{I}_n \uparrow I(G;X)$.

We call $\mathcal{I}_n$ the *attainable information* and $I(G;X)$ the *channel information*. The distinction is the practical content of this paper: $I(G;X)$ is what radiogenomics is ultimately about, but $\mathcal{I}_n$ is what a study of size n can ever demonstrate, and the two differ by an order of magnitude at realistic cohort sizes.

Corollary 2 (The quartic law: why weak directions are invisible)**.** *Since $n/(n+\nu) \le n/\nu$,*

$$\mathcal{R}_n(\rho^2)\ \le\ \min\bigg(\rho^2,\ \frac{n\,\rho^4}{d^\star(1-\rho^2)}\bigg). \tag{26}$$

In the weak-channel regime $\rho^2 \ll d^\star/n$ the attainable recoverability decays like ρ^4, not ρ^2: halving a canonical correlation quarters what a finite cohort can extract from it.

Corollary 2 is the formal version of a familiar disappointment. A genomic direction with a real but modest channel correlation — say $\rho^2 = 0.05$ — is not merely five times weaker than one at $\rho^2 = 0.25$; at $n = 200$ and $d^\star = 20$ it is attainable at $\mathcal{R}_n = 0.017$ versus 0.192, a factor of eleven. The consequence for study design is that radiogenomics at realistic n can see only the few strongest axes of the channel, and effort spent widening the gene panel buys directions that are, by (26), precisely the ones a cohort of that size cannot certify.

Corollary 3 (Concentration maximizes attainable information)**.** *The per-direction contribution $u \mapsto -\frac{1}{2}\log(1-\mathcal{R}_n(u))$ is convex on $[0,1]$. Consequently, among all spectra with a fixed total signal $T = \sum_i \rho_i^2$ and a per-direction cap $\rho_i^2 \le \bar{\rho}^2$, the attainable information* (25) *is maximized by* concentration*: fill $m = \lfloor T/\bar{\rho}^2 \rfloor$ directions at the cap and place the remainder $T - m\bar{\rho}^2$ in one more. Writing $g(u) = -\frac{1}{2}\log(1 - \mathcal{R}_n(u))$,*

$$\mathcal{I}_n \;\le\; m\,g(\bar{\rho}^2) \;+\; g\big(T - m\bar{\rho}^2\big)\,. \tag{27}$$

Corollary 3 is what converts a bound on a *single* axis into a bound on total information retention, and it matters because the two are easy to conflate. A limit on ρ_1^2 — the maximum over canonical axes, which is what a rank-one calibration delivers — bounds only the leading direction's contribution; since $\mathcal{I}_n$ sums over up to $\min(p^\star, d^\star)$ directions, that figure is a *lower* bound on the total, not an upper one. Equation (27) closes the gap given a total-signal budget.

One caveat is intrinsic rather than technical: T itself is not boundable from data, because signal spread thinly enough across many directions is undetectable at any n. That is precisely the regime in which convexity makes each contribution negligible, which is why Section 20.3 bounds the information directly — taking a supremum over spectra consistent with the observed statistic — rather than bounding T and then converting.

Corollary 4 (Optimal imaging feature budget)**.** *Enlarging the imaging representation cannot lower the population spectrum (data processing, Section 11), but raises every learning cost ν_i in proportion to $d^\star$. The attainable information is therefore maximized at a finite working dimension,*

$$d^\star_{\mathrm{opt}}(n) \;=\; \arg\max_{d^\star}\; \mathcal{I}_n\big(\{\rho_i^2(d^\star)\},\, d^\star\big), \tag{28}$$

and the envelope $U(n) = \max_{d^\star} \mathcal{I}_n$ is the ceiling once the analyst is allowed to choose the representation.

Corollary 4 puts the working-dimension rule of Section 13.5 — chosen there on identifiability grounds — on a decision-theoretic footing: $p^\star, d^\star \le n/\tau$ is a heuristic for the same trade-off that (28) optimizes exactly.

Binary genomics. For mutation calls the ceiling composes with the probit link of Section 15.3: substituting the attainable latent recoverability $\mathcal{R}_n(R_j^Z)$ for R_j^Z in (35) gives the AUC any classifier trained on n patients can reach for gene j,

$$\mathrm{AUC}_j(n) \;\le\; \Phi\left(\sqrt{\frac{\mathcal{R}_n(R_j^Z)}{2\,\big(1 - \mathcal{R}_n(R_j^Z)\big)}}\right). \tag{29}$$

This is the object to compare against a reported per-gene AUC, and it explains why AUCs in the 0.6–0.7 band are the norm in this literature rather than a failure of modeling.

12.3 What the ceiling does and does not assume

Three caveats delimit (23), and each is informative.

(i) It is an average-case bound. The prior (22) is the model's own symmetry, not an extra assumption about the data, but the resulting statement is about the risk averaged over signal directions. For a *particular* β with exploitable extra structure — genuine sparsity in the imaging basis, say — an estimator that knows the structure can exceed (23). This makes the bound a diagnostic rather than a wall: a model that reproducibly clears $\mathcal{R}_n$ is evidence of structure the linear–Gaussian channel does not encode, and is worth investigating rather than dismissing. Empirically (Section 12.4 and the three cohorts) we never observed such a crossing.

(ii) It is generous to the learner. The bound grants oracle knowledge of ρ^2, Σ_X and the noise level; a real analyst estimates all three, so achievable performance is strictly below (23). It also charges nothing for estimating the genomic target direction v itself, which in practice is another $p^\star$-dimensional estimation problem.

(iii) It inherits the model's misspecification. $\mathcal{R}_n$ is exact under the linear–Gaussian law. Section 13.7 quantifies the departure in each cohort; as there, the copula transform secures the marginals and the residual joint non-Gaussianity is signed conservatively.

12.4 Simulation: models approach the ceiling but do not cross it

Proposition 7 makes a two-sided prediction that is straightforward to falsify: no trained model should exceed $\mathcal{R}_n$, and the best one should track it, or the bound is vacuous. We test both on a planted linear–Gaussian channel with known spectrum $\{\rho_i^2\} = \{0.60, 0.35, 0.15, 0.05\}$ and $d^\star = 20$, sweeping the training size over $n \in [20, 4000]$ and, at each size, fitting six model families — ordinary least squares, cross-validated ridge, RBF kernel ridge, a random forest, gradient boosting, and a two-layer neural network — to predict the leading genomic canonical score from imaging, scored on 1.2×10^4 held-out patients and averaged over 40 replicates with independently drawn signal directions (Fig. 4).

Both predictions hold. Across all 84 model-by-n cells no family exceeds $\mathcal{R}_n$ by more than Monte-Carlo error (two cells at $n = 4000$ sit 2.1 standard errors above, against 1.9 expected by chance; a dedicated high-precision rerun at that cell with 200 replicates and a 6×10^4 test set returns 0.5979 ± 0.0002 against the ceiling 0.5980). Ridge tracks the ceiling to within 0.01 from $n = 95$ onward, so the bound is tight; the flexible families sit below it, since nonlinearity buys nothing against a linear–Gaussian channel and costs variance. The in-sample R^2 of the same OLS fit, by contrast, exceeds the *channel* value ρ_1^2 for all $n \lesssim 100$ and diverges to 1 as $n \to d^\star$ — the failure mode that Section 13.5 diagnoses, now measured against the correct benchmark.

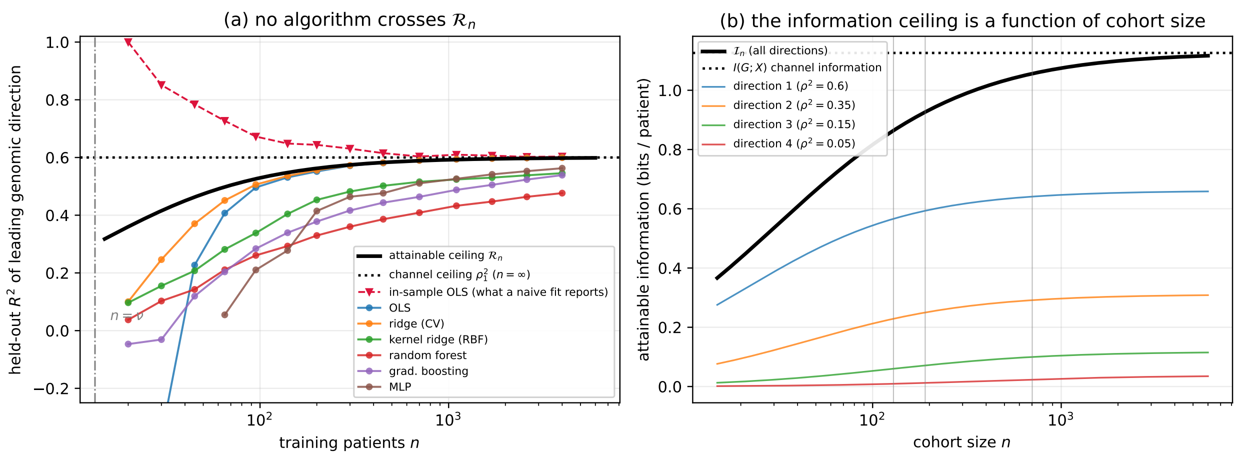


Figure 4: The attainable ceiling on a planted linear–Gaussian channel. **(a)** Held-out R^2 of six model families versus training size, against $\mathcal{R}_n$ (black) and the channel value $\rho_1^2 = 0.60$ (dotted). No family crosses the ceiling; cross-validated ridge attains it. In-sample OLS (red) sits above the channel value at small n. The dash-dotted line marks $n = \nu = 13.3$, where half the channel recoverability becomes attainable. **(b)** Attainable information $\mathcal{I}_n$ in bits, total and per-direction, with the three cohort sizes of Sections 17–19 marked. Weak directions contribute almost nothing at attainable n, as Corollary 2 requires.

12.5 Estimating the ceiling from data

Equation (23) is a function of the unknown channel value ρ^2, so reporting it requires an estimate. Two routes, both used below.

Learning-curve fit. $\mathcal{R}_n$ is a one-parameter family in n. Subsampling a cohort to a grid of training sizes, recording the best held-out R^2 at each, and fitting (23) by least squares returns an estimate $\hat{\rho}^2$ of the *channel* value together with a test of the functional form: if the observed learning curve is not hyperbolic in n with the predicted curvature, the model is wrong. This also extrapolates — it answers "what would this cohort give at $n = 10^4$?" — which a single-n cross-validated number cannot.

Test inversion. Because $\mathcal{R}_n$ is strictly increasing in ρ^2 (its derivative is $n\rho^2[2d^\star + \rho^2(n - d^\star)]$ over a positive square, hence positive for all $0 < \rho^2 \leq 1$), a one-sided upper confidence limit $\bar{\rho}^2$ for the channel value yields a valid one-sided upper confidence limit $\mathcal{R}_n(\bar{\rho}^2)$ for the ceiling. We take $\bar{\rho}^2$ from the semi-parametric calibration of Section 17.4, in which signal of known strength is planted into permuted real data so that the reference distribution at $\rho^2 = 0$ is exactly the permutation null. This is the route to the headline "$\leq$" statements.

13 Estimation from finite samples

In applications p and d may rival or exceed n, the noise is not isotropic, and A, Σ_G, σ^2 are unknown. Because Propositions 4 and 5 require only $\Sigma_G, \Sigma_X, \Sigma_{GX}$, we estimate those directly. Figure 5 summarizes the full pipeline, from the two feature matrices through regularized covariances, the whitened cross-modal CCA matrix, its SVD, and the resulting recoverability spectrum and image-identifiable genomic directions.

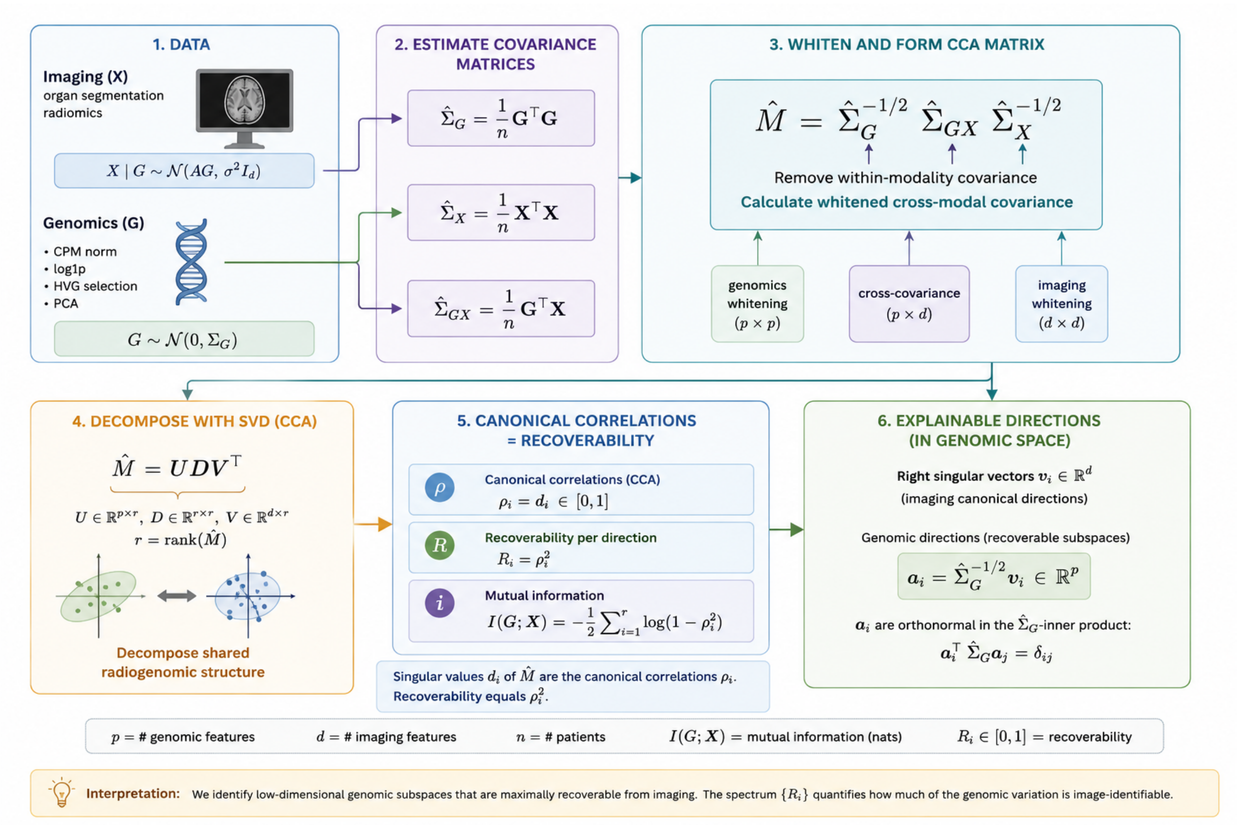


Figure 5: Overview of the recoverability estimator. From the imaging (X) and genomic (G) feature matrices we form regularized covariances $\hat{\Sigma}_G, \hat{\Sigma}_X, \hat{\Sigma}_{GX}$, whiten and assemble the cross-modal matrix $\hat{M} = \hat{\Sigma}_G^{-1/2}\hat{\Sigma}_{GX}\hat{\Sigma}_X^{-1/2}$, and take its SVD. The singular values are the canonical correlations $\hat{\rho}_i$; the recoverabilities are $\hat{R}_i = \hat{\rho}_i^2$ and the mutual information is $I(G;X) = -\frac{1}{2}\sum_i \log(1-\hat{\rho}_i^2)$; the left singular vectors map back to image-identifiable genomic directions $\hat{a}_i$. The spectrum $\{\hat{R}_i\}$ quantifies how much of the genomic variation is identifiable from imaging and along how many directions.

13.1 Regularized covariances

Replace sample covariances by shrinkage estimators (e.g. Ledoit–Wolf [36]), $\hat{\Sigma}_G = (1-\alpha_G)\,S_G + \alpha_G\,\nu_G I_p$ and likewise $\hat{\Sigma}_X$, with $S_G = n^{-1}\mathbf{G}^\top\mathbf{G}$ the centered sample covariance; the cross-covariance is $\hat{\Sigma}_{GX} = n^{-1}\mathbf{G}^\top\mathbf{X}$. Shrinkage keeps $\hat{\Sigma}_G, \hat{\Sigma}_X$ invertible and well-conditioned, which is required for the whitening in (12). When p is very large it is practical to first project genomics onto its top principal components (or onto curated pathway scores), reducing p to a manageable $p' \ll n$ before estimation; the rank limit of Proposition 6 means little is lost provided p' exceeds the recoverable dimension.

Gaussian-copula marginals. The model's Gaussian prior constrains the *marginals* of each feature, not only their dependence. Radiomic descriptors in particular are badly non-Gaussian — heavy-tailed shape and texture statistics with excess kurtosis in the tens (Section 17) — so the raw second moments $\hat{\Sigma}_G, \hat{\Sigma}_X$ are dominated by a few outlying patients and the whitening in (12) is unstable. A simple and assumption-faithful remedy is the *Gaussian-copula marginal* *transform*: replace each feature by its rank mapped through the inverse normal CDF, $\tilde{M}_{ij} = \Phi^{-1}\big(\mathrm{rank}_i(M_{\cdot j})/(n+1)\big)$, so every marginal is exactly standard normal and only the dependence structure (hence the canonical correlations) survives. This is the continuous-feature analog of the latent-Gaussian construction of Section 15 and is the default applied to both modalities in the analyses below.

13.2 Reduced-rank and ridge regression

The transfer map, if needed explicitly, is the multivariate regression of X on G. Ridge gives $\hat{A}^\top = (\mathbf{G}^\top\mathbf{G} + \gamma I_p)^{-1}\mathbf{G}^\top\mathbf{X}$, and $\hat{\sigma}^2$ follows from the residual covariance. Imposing $\mathrm{rank}(A) \le k$ (reduced-rank regression) bakes Proposition 6 into the fit and is equivalent, up to regularization, to truncating the SVD of M at k components.

13.3 Regularized CCA

The estimator used in the companion notebook solves (18) with the regularized covariances of Section 13.1: form $\hat{M} = \hat{\Sigma_G}^{-1/2}\hat{\Sigma_{GX}}\hat{\Sigma_X}^{-1/2}$, take its SVD $\hat{M} = \hat{U}\hat{D}\hat{V}^\top$, and read off $\hat{\rho}_i = \hat{d}_i$ (clipped to $[0,1)$), $\hat{R}_i = \hat{\rho}_i^2$, genomic directions $\hat{a}_i = \hat{\Sigma_G}^{-1/2}\hat{u}_i$, imaging directions $\hat{b}_i = \hat{\Sigma_X}^{-1/2}\hat{v}_i$. The Bayes-optimal predictor and posterior covariance follow from (7)–(8) with hats throughout. This is exactly regularized CCA — the recoverability program and CCA are the same computation.

13.4 Probabilistic CCA: the latent-variable view

The rank limit motivates an explicit low-dimensional generative form. With a shared latent $Z \in \mathbb{R}^k$,

$$Z \sim \mathcal{N}(0, I_k), \qquad G = W_G Z + \mu_G + \epsilon_G, \qquad X = W_X Z + \mu_X + \epsilon_X, \tag{30}$$

$\epsilon_G \sim \mathcal{N}(0, \Psi_G)$, $\epsilon_X \sim \mathcal{N}(0, \Psi_X)$ independent (probabilistic CCA; 30). Here k *is* the dimension of the image-identifiable genomic subspace, Z is the shared biological state, and ϵ_G, ϵ_X are modality-private variation. The implied moments are $\Sigma_G = W_G W_G^\top + \Psi_G$, $\Sigma_X = W_X W_X^\top + \Psi_X$, $\Sigma_{GX} = W_G W_X^\top$; the maximum-likelihood loadings are the canonical directions scaled by the canonical correlations, recovering Section 13.3. The model is fit by EM, which also yields posterior latent estimates $\mathbb{E}[Z \mid x]$ and naturally accommodates anisotropic (per-feature) noise, unlike the isotropic $\sigma^2 I_d$ of the base model.

13.5 Sample-size regimes and dimensional collapse

Equations (7)–(18) are population identities; the estimators of Sections 13.1–13.4 inherit a finite-sample bias whose magnitude is governed entirely by the aspect ratios p/n and d/n. The bias is not benign, and below a critical sample size the recoverability spectrum breaks down in a specific, predictable way.

Three regimes. Let $p^\star, d^\star$ be the working dimensions after any pre-reduction (e.g. PCA), and write $\kappa_G = p^\star/n$, $\kappa_X = d^\star/n$.

1. *Classical* ($\kappa_G + \kappa_X \ll 1$). Sample CCA is consistent; $\hat{\rho}_i \to \rho_i$ at the usual $n^{-1/2}$ rate and the in-sample spectrum is a faithful estimate of the population one.
2. *High-dimensional* ($\kappa_G + \kappa_X \lesssim 1$, both bounded away from 1). Under the null $\Sigma_{GX} = 0$, the squared sample canonical correlations follow a Wachter-type law on $[(\sqrt{\kappa_G(1-\kappa_X)} - \sqrt{\kappa_X(1-\kappa_G)})^2, (\sqrt{\kappa_G(1-\kappa_X)} + \sqrt{\kappa_X(1-\kappa_G)})^2]$ [33, 34]; the leading sample value already sits well above zero by geometry. Estimates of ρ_i are systematically inflated and ordinary CCA must be replaced by its regularized form, with shrinkage parameters tuned by cross-validation.
3. *Saturating* ($p^\star + d^\star \ge n$). The rows of $\mathbf{G}$ and $\mathbf{X}$ live in subspaces whose intersection has dimension at least $p^\star + d^\star - n + 1$; consequently $\hat{\rho}_1 = \cdots = \hat{\rho}_q = 1$ for $q = \min(p^\star, d^\star) - (n - \max(p^\star, d^\star))_+$ *regardless of any underlying association*. The empirical mutual information diverges, the in-sample $\mathrm{tr}(\hat{\Sigma_{GX}}\hat{\Sigma_X}^{-1}\hat{\Sigma_{XG}})/\,\mathrm{tr}(\hat{\Sigma_X})$ approaches the rank fraction of the genomic span, and the permutation null collapses onto the observed value (its p-values concentrate near 1). Nothing in the spectrum is estimable in this regime without shrinkage *plus* dimension reduction; shrinkage alone keeps the matrices invertible but does not restore identifiability.

Quantitative warning signs. Three diagnostics, jointly, separate real signal from dimensional artifact:

- the gap between in-sample $\hat{R}_i$ and the K-fold value $\hat{R}_i^{\mathrm{cv}}$; under regime 3 this gap approaches $\hat{R}_i - 0$;
- the permutation null for $\hat{\rho}_1$: if the observed value lies inside the bulk, the result is consistent with chance;
- the cross-validated R^2 of the leading canonical score $a_1^\top g$ predicted from x on a held-out split; in regime 3 it is typically large and *negative* (worse than predicting the mean), a behavior seen empirically on TCGA-KIRC at $n = 131$ with $p^\star = 100, d^\star = 107$ in the companion notebook.

Recommendations. Choose $p^\star$ and $d^\star$ *adaptively from* n, not from the native modality dimensions:

- Target $p^\star + d^\star \le n/\tau$ with $\tau \ge 5$ as a working rule (we use $p^\star, d^\star \le \lfloor n/\tau \rfloor$ per side in the notebook; $\tau \in [5, 10]$ trades bias for resolution). Reduce both modalities by PCA when needed — reducing only the larger side leaves the estimator in regime 3.
- Treat $\hat{R}_i^{\mathrm{cv}}$ (Section 13.6) as the primary estimand and report the in-sample $\hat{R}_i$ only with the gap made explicit; the in-sample fraction of imaging variance attributable to genomics should be reported alongside a cross-validated counterpart, e.g. $\sum_i \hat{R}_i^{\mathrm{cv}} / \min(p^\star, d^\star)$.
- Power planning. The leading population correlation ρ_1 is detectable above the Wachter bulk only when its squared value exceeds $(\sqrt{\kappa_G(1-\kappa_X)} + \sqrt{\kappa_X(1-\kappa_G)})^2$; solving for n gives a minimum sample size given a hypothesized effect. For $\rho_1^2 = 0.2$ and a balanced $p^\star = d^\star$, this requires roughly $n \gtrsim 25\, p^\star$, an order of magnitude beyond what cohort sizes in radiogenomics usually offer — reinforcing the need for aggressive pre-reduction.
- Probabilistic CCA (Section 13.4) with a small, fixed latent dimension k is the most parsimonious option when n is the binding constraint; rank-k regularization sidesteps the high-dimensional bulk by construction.

The practical message is conservative: in radiogenomic cohorts, sample size is almost always the binding constraint and a defensible analysis is the one whose effective $p^\star + d^\star$ is a small fraction of n, with both sides reduced symmetrically and every reported number a cross-validated or permutation-calibrated one.

Figure 6 illustrates these regimes on the synthetic instance, where the planted truth is known. As n grows with $p^\star, d^\star$ scaled to keep $\kappa_G + \kappa_X$ bounded, the in-sample $\hat{\rho}_1$ sits at or near 1 across the small-n saturating regime — indistinguishable from the permutation null — while the cross-validated $\hat{R}_1^{\mathrm{cv}}$ only separates from zero once n clears the Wachter edge. The empirical mutual information shows the same threshold behavior.

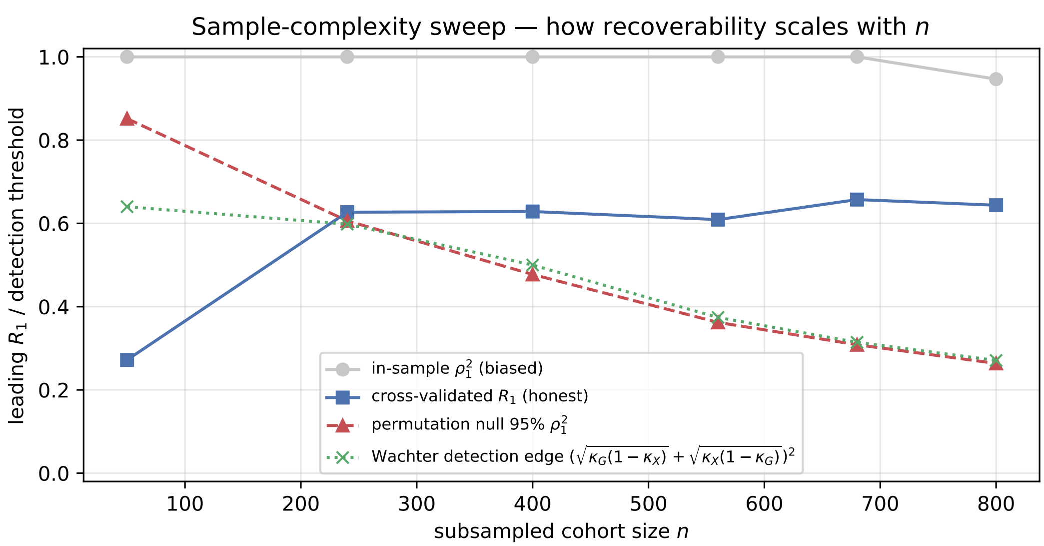


Figure 6: Sample-complexity sweep on the synthetic instance with known truth. In-sample $\hat{\rho}_1$ tracks the null until n exceeds the Wachter edge; cross-validated $\hat{R}_1^{\mathrm{cv}}$ and empirical $\hat{I}(G; X)$ only become informative past that threshold. Compare to Figure 13c for the same diagnostic on TCGA-KIRC.

The sweep above grows $p^\star, d^\star$ with n, which entangles "more data" with "bigger model". The complementary question — how much data is needed at a *fixed* model — is answered by Figure 7, which holds $p^\star = d^\star = 20$ and varies only n on the same synthetic instance, whose planted leading recoverability $\rho_1^2 = 0.65$ is known in closed form (Proposition 5). Two thresholds separate. The cross-validated *point* estimate $\hat{R}_1^{\mathrm{cv}}$ is already close to the truth at small n (within 90% by $n \approx 80$, i.e. $n \approx 2(p^\star + d^\star)$), because the planted signal is strong. But the estimate cannot be *certified* against the permutation null until later: the null 95% quantile is inflated to ≈ 1 in the saturating regime ($n \lesssim p^\star + d^\star$) and only decays below the estimate at $n \approx 176 \approx 4.4(p^\star + d^\star)$. The practical reading is that "how much data" depends on the claim: recovering the right number takes far less data than proving it is not an artifact of the high-dimensional geometry, and the cohort downsample sweeps (Sections 19.5) report the latter, conservative threshold.

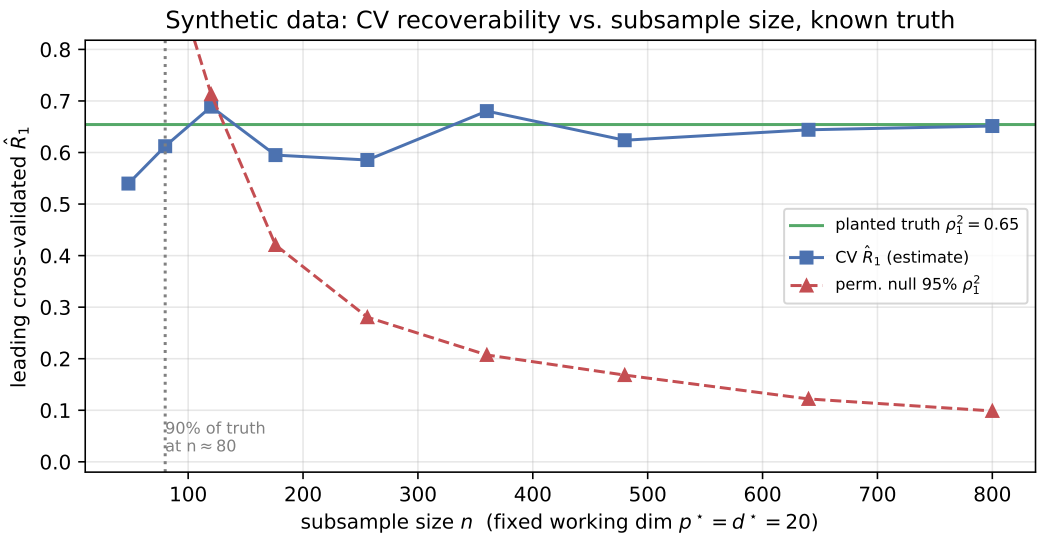


Figure 7: Fixed-model downsampling sweep on the synthetic instance ($p^\star = d^\star = 20$, planted $\rho_1^2 = 0.65$). The cross-validated estimate (blue) tracks the known truth (green) even at small n, but the permutation null 95% quantile (red, dashed) is saturated near 1 in the small-n regime and only falls below the estimate at $n \approx 176$. Recovering the value needs less data than certifying it. Companion to the growing-model sweep of Fig. 6 and the cohort sweeps (Fig. 17e).

13.6 Honest recoverability: cross-validation and permutation

In-sample canonical correlations are biased upward, severely so when p, d are not small relative to n. Two safeguards, both implemented in the notebook:

- *Cross-validation.* Estimate $\hat{a}_i, \hat{b}_i$ on a training fold; on the held-out fold form the scores $\hat{a}_i^\top g$ and $\hat{b}_i^\top x$ and report their out-of-sample squared correlation as the honest $\hat{R}_i$. The gap to the in-sample value quantifies overfitting.
- *Permutation test.* Repeatedly permute the patient labels of one modality, refit, and collect the leading canonical correlation to build a null distribution; the permutation p-value tests whether any genomic direction is image-identifiable beyond chance.

Scale-matched effective rank. A subtlety arises when these two safeguards are combined to count *how many* directions are identifiable. The in-sample permutation null of the leading $\hat{\rho}_1$ is itself inflated toward 1 by the high-dimensional geometry of Section 13.5, whereas $\hat{R}_i^{\mathrm{cv}}$ is honestly deflated; thresholding the deflated statistic against the inflated null forces the estimated rank to zero regardless of any true signal. The two must be compared on the same scale. We therefore build the null of the *cross-validated* statistic directly: permute patient labels, recompute the full K-fold $\hat{R}_i^{\mathrm{cv}}$, and define the *effective image-identifiable rank* as the number of directions whose observed $\hat{R}_i^{\mathrm{cv}}$ exceeds its own permutation 95% quantile at permutation $p < 0.05$. This is the estimand we report below.

Figure 8 shows both diagnostics on the synthetic instance. The in-sample $\hat{R}_i$ overshoots the planted truth by the amount predicted by the high-dimensional bias, while the 5-fold $\hat{R}_i^{\mathrm{cv}}$ tracks the truth for identifiable directions and collapses toward the null 95% quantile thereafter. Figure 9 shows the permutation null on $\hat{\rho}_1$: when the true signal is nonzero, the observed value sits cleanly outside the null bulk and the per-direction p-values fall below 0.01 for the top three components, matching the planted rank.

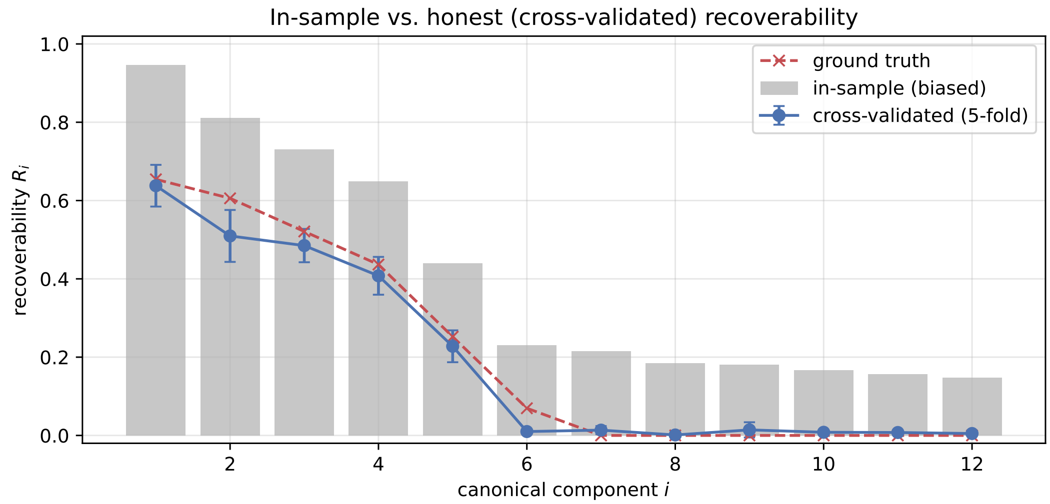


Figure 8: Cross-validated recoverability on a synthetic instance with planted rank $k = 6$ (the sixth correlation near zero). In-sample $\hat{R}_i$ (gray) is biased upward; 5-fold $\hat{R}_i^{\mathrm{cv}}$ (blue) tracks the planted truth for the identifiable directions and collapses to the null floor afterwards.

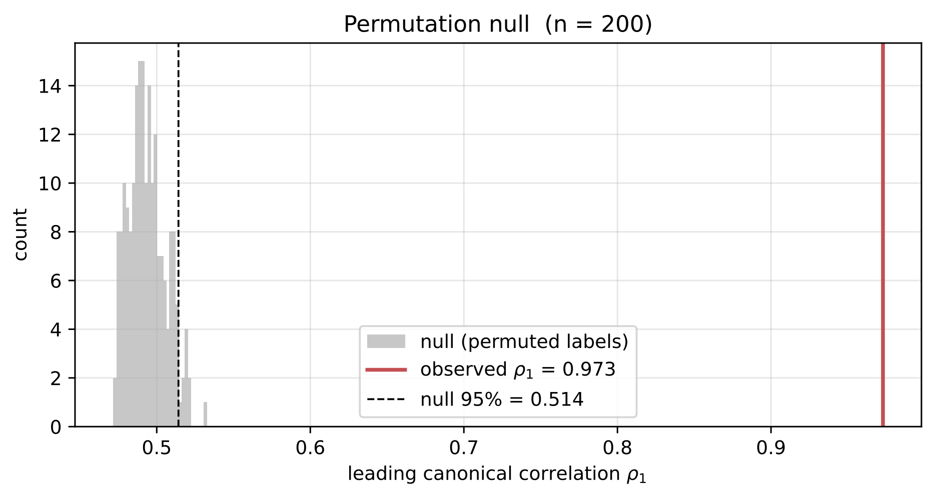


Figure 9: Permutation null on the leading canonical correlation $\hat{\rho}_1$ for the synthetic instance. Red marks the observed value; $p < 0.01$ for the top three directions, recovering the planted rank.

The same optimism appears in the aggregate fraction of imaging variance attributable to genomics. Figure 10 shows the in-sample share $\sum_i \hat{R}_i / \min(p^\star, d^\star)$ alongside its cross-validated counterpart on the synthetic instance: the in-sample number sits well above the planted truth while the CV value tracks it. The same diagnostic on TCGA-KIRC (Section 17) reports 0.193 in-sample against 0.012 cross-validated — a sixteen-fold gap that is consistent with the synthetic mechanism rather than with any genuinely recoverable signal.

13.7 Checking the model assumptions

The closed-form recoverability and mutual information are exact *under* the linear–Gaussian law; before trusting the numbers we verify, upfront, how far each cohort departs from it. Three checks bracket the assumptions the estimator actually uses.

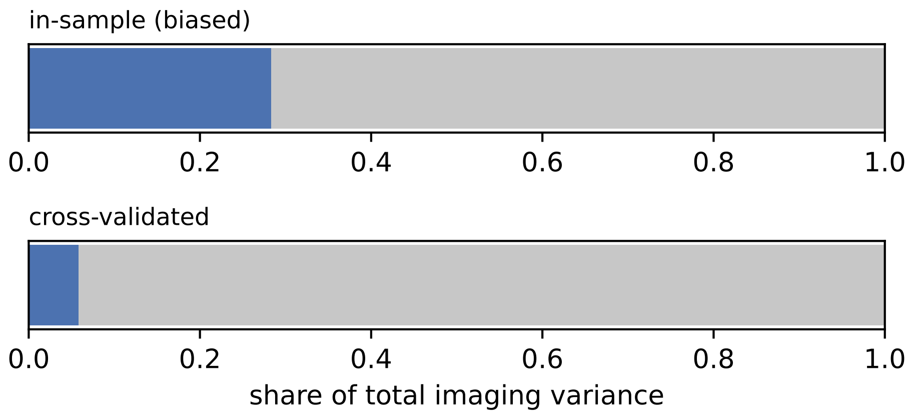


Figure 10: Share of imaging variance attributable to genomics on the synthetic instance: in-sample (gray) versus 5-fold cross-validated (blue), with the planted truth marked. The in-sample share is biased upward by the high-dimensional inflation described in Section 13.5.

(i) Marginal and joint Gaussianity. The Gaussian-copula transform (Section 13.1) makes every *marginal* exactly normal by construction; the question is whether the *joint* $[G_{\text{pca}}, X_{\text{pca}}]$ is. We test it with a Mahalanobis-distance QQ plot against the χ^2_p law (squared Mahalanobis distances are χ^2_p under joint MVN) and Mardia's standardized skew/kurtosis ratios (Fig. 11). Per-modality marginal normality does not imply joint normality, so this is the assumption the spectrum genuinely relies on.

(ii) Linearity of the dependence. Canonical correlation sees only second-order (linear) coupling. We complement it with the distance correlation [52], which is zero *iff* G_{pca} and X_{pca} are independent and therefore detects non-linear and higher-order dependence the linear–Gaussian estimator is blind to. We report a permutation p-value (the statistic has a positive finite-sample bias, so its null, not its raw value, is the reference). A significant distance correlation where the cross-validated recoverability is null would indicate a channel the linear bound understates.

(iii) Per-feature marginals. As documentation of the copula step, we summarize per-feature skewness and excess kurtosis on the raw features and confirm both vanish after the transform.

Findings. Table 1 collects the three checks across cohorts. Two patterns are robust. First, the copula transform is doing real work: raw genomic marginals are severely non-Gaussian (78% of KIRC probes fail $|\text{skew}| \leq 2$, $|\text{kurt}| \leq 7$, median excess kurtosis 70), and all go to 0% afterwards. Second, and less obviously, *marginal Gaussianity does not buy joint Gaussianity*: even post-copula, the share of samples beyond the χ^2 99% shell is 7–16% rather than the nominal 1%. Part is finite-sample (the ratio $n/(\dim G + \dim X) \approx 2.5$ in all three inflates the Mahalanobis tail), but it shrinks monotonically with n ($16\% \to 12\% \to 8\%$ for KIRC $\to$ NSCLC $\to$ ADNI), implicating genuine residual *tail dependence* the copula cannot remove. The Mardia skew ratio stays ≈ 0, so the joint is symmetric but leptokurtic. The distance correlation is significant in every cohort ($p = 0.005$) but only modestly above its inflated null, and its ordering tracks the *linear* recoverability rather than exceeding it — there is no cohort where non-linear dependence lights up while the linear channel is dead.

	TCGA-KIRC	NSCLC	ADNI
n / joint dim	$190/76$	$129/50$	$702/280$
Joint kurt. ratio (1 = MVN)	1.17	1.14	1.02
Tail beyond $\chi^2_{.99}$ (1% = MVN)	15.8%	12.4%	8.1%
Genomic marginals non-normal (raw $\to$ post)	$78\% \to 0\%$	$57\% \to 0\%$	$24\% \to 0\%$
Distance corr. (obs / null$_{95}$ / p)	$0.43/0.37/.005$	$0.34/0.31/.005$	$0.32/0.27/.005$

Table 1: Model-assumption diagnostics. The copula removes all marginal non-normality; the joint remains mildly heavy-tailed (worst in the small tumor cohorts, near-Gaussian in ADNI); the distance correlation reveals no non-linear channel beyond the linear one.

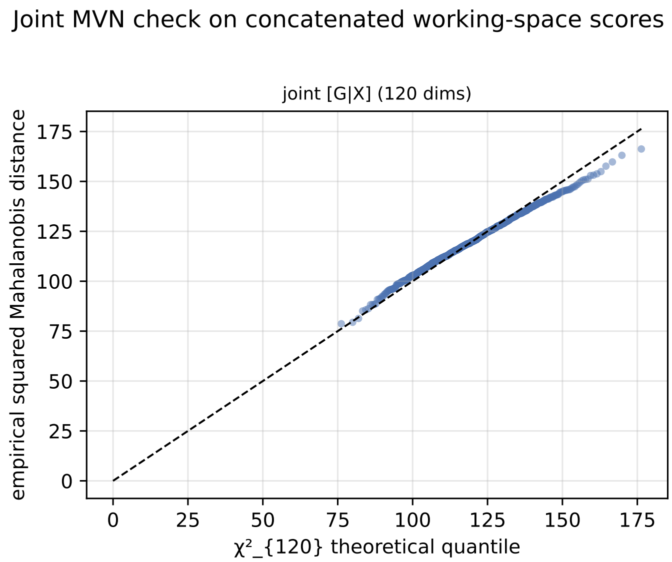
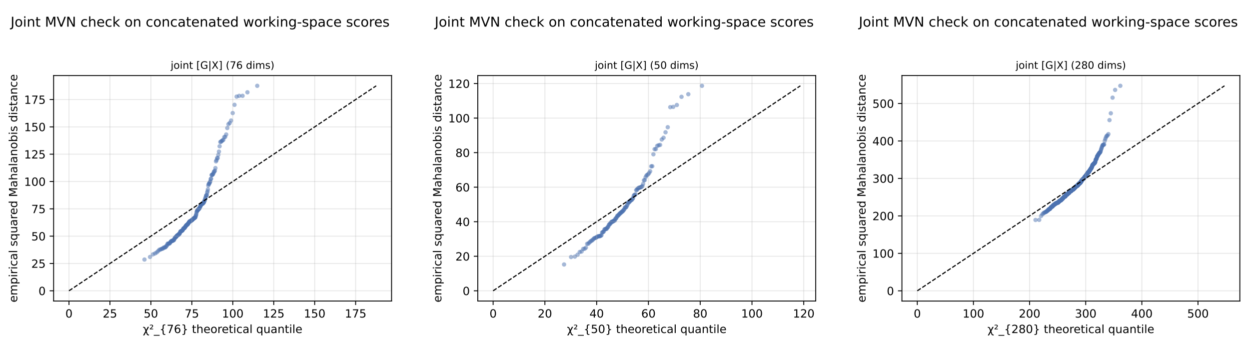


Figure 11: Joint-MVN check on the concatenated working-space scores $[G_{\text{pca}}, X_{\text{pca}}]$ for the synthetic instance: squared Mahalanobis distances against χ_p^2 quantiles lie on the line, with kurtosis ratio ≈ 1 and a 0.4% tail beyond the 99% shell. The real cohorts (Fig. 12) depart from this in their upper tails.



(a) TCGA-KIRC (b) NSCLC (c) ADNI

Figure 12: Joint-MVN QQ plots on the three real cohorts (squared Mahalanobis distances of $[G_{\text{pca}}, X_{\text{pca}}]$ vs. χ_p^2 quantiles; the line $y = x$ is exact joint Gaussianity). The copula makes every marginal Gaussian, so departures here are purely *joint* non-Gaussianity. All three track the line through the bulk and lift above it in the upper tail — the residual leptokurtosis (heavy joint tails) quantified in Table 1. The deviation is largest in the small tumor cohorts (a, b) and mildest in ADNI (c), the cohort with the most favorable $n/(\dim G + \dim X)$ ratio, consistent with the part of the inflation that is finite-sample. Together with the synthetic ideal (Fig. 11) these panels are the visual companion to the assumption table.

What this does and does not cost the upper bound. It is worth separating two claims that the phrase "$I(G; X)$ is an upper bound" can mean. The *operational* bound — that no estimator, linear or non-linear, can recover G from X better than $I(G; X)$ allows — is the data processing inequality $G \to X \to \hat{g}(X)$ and holds for the *true* joint law regardless of its shape (Section 15.2). What requires Gaussianity is the second, computational claim: that the *number* we report, $-\frac{1}{2}\sum_i \log(1 - \hat{\rho}_i^2)$, equals that true $I(G; X)$. The copula secures the marginals; the residual joint non-Gaussianity above means the two are equal only approximately. Crucially, the mis-specification is one-sided in a favorable direction: because canonical correlation captures only linear coupling, ignoring higher-order dependence can only make the Gaussian formula *under*-count the true $I(G; X)$, never over-count it — so the reported figure is conservative as a statement about the true channel, not optimistic. The distance-correlation check bounds how much is being missed (little), and the finite-sample optimism that runs the *other* way is removed separately by the permutation-calibrated upper confidence limit (Section 17.4). The net reading: significance and the identifiable-rank conclusions are distribution-free and stand; the MI *magnitudes* are best read as approximate, most trustworthy in ADNI (near-Gaussian joint, large n) and most caveated in the small, heavier-tailed tumor cohorts.

A signed bias, made precise. The favorable direction is not heuristic; the copula pins it down. Write $I(G;X) = h(G) + h(X) - h(G,X)$. The copula forces each marginal to be exactly Gaussian, so $h(G)$ and $h(X)$ equal their Gaussian values and the *only* term in which the true law can differ from the Gaussian surrogate is the joint entropy $h(G,X)$. Because the Gaussian is the maximum-entropy law for a fixed covariance, $h(G,X) \leq h_{\text{gauss}}(G,X)$, whence

$$I(G;X) - \left(-\tfrac{1}{2}\textstyle\sum_i \log(1-\rho_i^2)\right) \;=\; h_{\text{gauss}}(G,X) - h(G,X) \;\geq\; 0, \tag{31}$$

with equality *iff* the joint is Gaussian. So, given Gaussian marginals, the population canonical-correlation mutual information is a genuine *lower* bound on the truth — exact under joint Gaussianity, an under-estimate otherwise — and the gap is precisely the non-Gaussian (non-linear, higher-order) dependence that CCA cannot see, which is what the distance correlation probes. (Without the Gaussian-marginal condition the marginal negentropies re-enter (31) and the sign is no longer guaranteed; securing this clean direction is a concrete reason to apply the copula.) The two error sources therefore push in *opposite* directions: population mis-specification can only *lower* the estimate relative to the true $I(G;X)$, while finite-sample inflation of $\hat{\rho}_i$ *raises* it. The permutation-calibrated upper confidence limit (Section 17.4) controls the second; (31) and the distance-correlation check bound the first. A reported figure that is small after both controls is therefore a trustworthy statement that the channel is genuinely narrow, not an artifact of either bias.

14 Interpretation for radiogenomics

The model reframes the radiogenomic question [1–4]. Rather than testing whether CT predicts each gene — a high-dimensional, low-power multiple-testing problem — it asks for the *recoverability spectrum* $\{\rho_i^2\}$ and the associated genomic directions $\{a_i\}$. The outputs are: (1) a small number of image-identifiable genomic axes, with gene/pathway loadings that make them biologically legible; (2) a hard bound $\Sigma_{G|X}$ on what any predictor can achieve, per direction; (3) a total information budget $I(G;X)$; and (4) an honest, cross-validated estimate of each. The governing principle is that radiogenomics should target *image-identifiable low-dimensional genomic subspaces*, whose dimension is set by the biology–imaging channel and not by the gene panel size.

Section 12 adds the fifth and, for a practitioner, the most consequential output: (5) a ceiling $\mathcal{I}_n$ on how much of that budget a cohort of size n can actually claim. The four quantities above are properties of the biology–imaging channel; $\mathcal{I}_n$ is a property of the *study*. Reporting it changes what a null result means. A per-gene AUC of 0.62 is uninformative in isolation, but against an attainable ceiling of 0.68 it is close to everything the cohort had to give, whereas against a ceiling of 0.95 it is a failure of the predictor. Three practical rules follow directly:

- *Report bits, not just p-values.* $\mathcal{I}_n$ is on a scale a clinician can price — one bit is one perfectly measured balanced binary biomarker — and it is comparable across disease, modality and cohort.
- *Size the study from Eq.* (24)*, not from convention.* The learning cost $\nu = d^\star/\text{SNR}$ makes the imaging representation a design variable on the same footing as the patient count, and Corollary 4 says the best feature budget is finite.
- *Prefer a short, pre-specified gene panel.* By Corollary 2 the directions a wider panel adds are exactly the weak ones a finite cohort cannot certify, and empirically (Section 20.8) a four-gene panel carries most of what the one cohort with a robust channel can deliver.

15 A Bernoulli extension for mutation data via a Gaussian copula

The linear–Gaussian model of Section 3 is well matched to continuous, approximately log-normal genomic readouts: log-CPM gene expression, methylation β- or M-values, log-CNV ratios. It is plainly misspecified for somatic mutation calls, where each $G_j \in \{0,1\}$ indicates the presence of a non-silent variant in gene j and the marginal frequencies p_j are small, gene-specific, and highly heterogeneous. This section extends the framework to such data by replacing the Gaussian prior on G with a *Gaussian-copula Bernoulli* (equivalently, multivariate probit) prior, while preserving the canonical-correlation / mutual-information machinery of Sections 7–8.

15.1 The latent-Gaussian generative model

Let $p_j = \Pr(G_j = 1)$ and $\tau_j = \Phi^{-1}(1-p_j)$, where Φ is the standard normal CDF. Introduce a latent vector

$$Z \sim \mathcal{N}(0, \Sigma_Z), \qquad \operatorname{diag}(\Sigma_Z) = I_p,$$

and define the observed mutation indicator by thresholding:

$$G_j = \mathbb{1}\{Z_j > \tau_j\}, \qquad j = 1, \dots, p.$$

By construction the marginals are correct, $G_j \sim \mathrm{Bern}(p_j)$, and the pairwise joint laws are governed by the tetrachoric correlation $\Sigma_{Z,jk}$, which can be any positive-definite correlation matrix — providing arbitrary dependence structure between mutated genes (e.g. mutual exclusivity, co-occurrence within pathways).

Imaging is coupled to the *latent* state, not to the indicators directly:

$$X \mid Z \sim \mathcal{N}\big(B\,Z,\ \sigma^2 I_d\big), \qquad B \in \mathbb{R}^{d\times p}. \tag{32}$$

This is the natural specification: the biological mechanism that produces an imageable phenotype (proliferation, vascularity, necrosis) responds to an underlying continuous burden, of which an observed mutation is a coarse, binarized readout. The joint structure factors as

$$Z \to G, \qquad Z \to X,$$

so G and X are conditionally independent given Z.

15.2 Mutual information and a tight DPI bound

Because G is a deterministic, memoryless function of Z, the data processing inequality yields the upper bound

$$I(G;X) \ \le\ I(Z;X), \tag{33}$$

with equality only if Z can be recovered from G (impossible whenever any $p_j \in (0,1)$, since thresholding is lossy). The right-hand side reduces exactly to the Gaussian formula of Section 7: if $\{\rho_i\}_{i=1}^k$ denote the canonical correlations of (Z, X) under (32), then

$$I(Z;X) \ =\ -\tfrac{1}{2}\sum_{i=1}^{k}\log\big(1-\rho_i^2\big), \qquad k = \mathrm{rank}\big(\Sigma_Z^{1/2} B^\top \Sigma_X^{-1} B \Sigma_Z^{1/2}\big), \tag{34}$$

with $\Sigma_X = B\Sigma_Z B^\top + \sigma^2 I_d$. The recoverability spectrum, the directions $\{a_i\}$, and the eigenproblem of Section 10 all transfer verbatim, with Σ_G replaced by Σ_Z.

The binarization gap. The slack in (33) is

$$I(Z;X) - I(G;X) \ =\ H(Z \mid G) - H(Z \mid G, X) \ =\ I(Z; X \mid G),$$

the residual information about Z not already revealed by the mutation call. For a single gene with frequency p_j, the entropy lost to binarization is $H(Z_j) - H(G_j) = \frac{1}{2}\log(2\pi e) - h_2(p_j)$ where h_2 is the binary entropy in nats; rare mutations ($p_j \ll 1/2$) are highly lossy, common ones near $p_j = 1/2$ less so. Thus (33) is tightest for cohorts dominated by intermediate-frequency mutations and loosest for rare-variant panels — a useful guide to when the bound is informative.

15.3 Per-gene recoverability and AUC ceilings

For a named mutation G_j, the Bayes-optimal predictor from X is the posterior probability

$$\pi_j(x) \ =\ \Pr(G_j = 1 \mid X = x) \ =\ \Phi\!\left(\frac{\mu_j(x) - \tau_j}{s_j}\right),$$

where $(\mu_j(x), s_j^2) = (\mathbb{E}[Z_j \mid X = x],\ \mathrm{Var}(Z_j \mid X = x))$ are the linear-Gaussian posterior moments for the latent coordinate, computed from (Σ_Z, B, σ^2) exactly as in Section 6. The *latent recoverability* $R_j^Z = \mathrm{Var}(\mathbb{E}[Z_j \mid X])/\mathrm{Var}(Z_j)$ is the direct analog of $R(v)$ in the Gaussian case, and the binary AUC of any predictor $\hat{G}_j(x)$ is bounded by

$$\mathrm{AUC}_j \ \le\ \Phi\!\left(\sqrt{\tfrac{R_j^Z}{1-R_j^Z}}\Big/\sqrt{2}\right), \tag{35}$$

attained by the Bayes classifier $\pi_j(x) > p_j$. Equation (35) is the natural object to compare against the empirical per-gene $\mathrm{AUC}_{\mathrm{LR}}$ and $\mathrm{AUC}_{\mathrm{RF}}$ of Section 17.7: an empirical AUC near 0.5 is consistent with either a small R_j^Z *or* a degenerate p_j, and (35) disentangles the two.

15.4 Estimation

The estimator splits into three independent pieces, each well-studied:

(i) **Marginal thresholds.** Estimate $\hat{p}_j = \bar{G}_{\cdot j}$ and set $\hat{\tau}_j = \Phi^{-1}(1 - \hat{p}_j)$. Apply a continuity correction or empirical-Bayes shrinkage for rare mutations.

(ii) **Latent correlation $\hat{\Sigma}_Z$.** Estimate pairwise tetrachoric correlations from $2{\times}2$ tables of (G_j, G_k) by maximum likelihood under the bivariate-normal copula, then project the resulting matrix to the nearest positive-definite correlation matrix (e.g. Higham's `nearcorr` [53]). For high-dimensional panels apply Ledoit–Wolf shrinkage as in Section 13.1.

(iii) **Cross-modal covariance $\hat{\Sigma}_{ZX}$.** The point-biserial covariance $\mathrm{Cov}(G_j, X_\ell)$ is related to the latent covariance by

$$\mathrm{Cov}(G_j, X_\ell) \;=\; \phi(\tau_j)\;\mathrm{Cov}(Z_j, X_\ell),$$

where ϕ is the standard normal density. Estimate $\mathrm{Cov}(G_j, X_\ell)$ from the sample and rescale by $\phi(\hat{\tau}_j)$ to obtain $\hat{\Sigma}_{ZX,j\ell}$.

With $(\hat{\Sigma}_Z, \hat{\Sigma}_{ZX}, \hat{\Sigma}_X)$ in hand, the regularized CCA of Section 13.3, the permutation null of Section 13.6, and the empirical MI bound (estimated, e.g., by k-nearest-neighbor or neural estimators [44, 45, 54]) carry over without further change. The dimensional-collapse warnings of Section 13.5 are if anything *more* acute, since tetrachoric estimation has higher variance than Pearson correlation, particularly for rare-rare gene pairs.

15.5 Mixed-modality genomics

In practice a single cohort carries both continuous (expression, CNV) and binary (mutation) features. The copula formulation accommodates this by partitioning the latent vector as $Z = (Z^{\mathrm{cts}}, Z^{\mathrm{bin}})$ with a single joint Σ_Z. The continuous block uses the identity link ($G^{\mathrm{cts}} = Z^{\mathrm{cts}}$ up to marginal standardization) and the binary block uses the probit link of Section 15.1. Estimation of Σ_Z proceeds blockwise — Pearson within the continuous block, tetrachoric within the binary block, polyserial across blocks — after which the rest of the pipeline is identical. This is the route by which the present manuscript's gene-expression analysis (Section 17) and the per-gene mutation validation (Section 17.7) can be unified into a single coherent recoverability calculation.

16 Anchor genes: how much of the channel is a handful of genes?

The ceiling says imaging can support only a few genomic directions at realistic n. It does not say *which*. A radiologist's operational question is sharper: if the recoverable subspace is small, is it spanned by genes we already know about?

Let $\mathcal{A} \subset \{1, \ldots, p\}$ be a hand-specified anchor panel — the canonical drivers and lineage markers of the disease. The chain rule splits the channel information exactly,

$$I(G; X) \;=\; I(G_{\mathcal{A}}; X) \;+\; I(G_{\bar{\mathcal{A}}}; X \mid G_{\mathcal{A}}), \tag{36}$$

and under the joint Gaussian law the conditional term is an ordinary mutual information between residualized variables: with $G^{\perp}_{\bar{\mathcal{A}}} = G_{\bar{\mathcal{A}}} - \Sigma_{\bar{\mathcal{A}}\mathcal{A}}\Sigma_{\mathcal{A}}^{-1}G_{\mathcal{A}}$ and $X^{\perp} = X - \Sigma_{X\mathcal{A}}\Sigma_{\mathcal{A}}^{-1}G_{\mathcal{A}}$,

$$I(G_{\bar{\mathcal{A}}}; X \mid G_{\mathcal{A}}) = I\big(G^{\perp}_{\bar{\mathcal{A}}};\, X^{\perp}\big), \tag{37}$$

computable by the same CCA machinery. Operationally this is Gram–Schmidt in the Σ_G metric: the genomics matrix is whitened *against the anchors*, so that "new information" means new relative to the panel rather than merely correlated with it. Without this step the pervasive co-expression structure of the transcriptome would credit the residual block with information the anchors already carry.

Definition 3 (Anchor sufficiency)**.** $\eta(\mathcal{A}) := I(G_{\mathcal{A}}; X)\,/\,I(G; X) \in [0, 1]$*, with the finite-sample counterpart* $\eta_n(\mathcal{A})$ *obtained by replacing both terms with their attainable versions* (25).

Two features of η_n deserve emphasis. First, it is the right currency for the question: a panel that captures 80% of the *channel* but requires 10^4 patients to fit is not a useful panel, and η_n says so. Second, the finite-sample version *favors* small panels, because the residual block pays learning cost for every direction it opens while the anchors pay for few. Combined with Corollary 2, this yields the design implication we test in Section 20.8: at realistic cohort sizes, restricting attention to a short list of established genes is not merely more interpretable than a transcriptome-wide search — it is closer to statistically optimal.

16.1 Estimation of the anchor split

The anchor block is used at its native resolution (one coordinate per gene, copula-transformed), the residual block and imaging are reduced to $d^\star$ principal components, and both terms of (36) are evaluated as attainable information (25) from K-fold held-out recoverability, so neither term is inflated by the in-sample bias of Section 13.5. The null for the residual term is built by permuting patient labels and recomputing the whole pipeline, residualization included; this is the test of whether *anything* survives the panel that a cohort of that size could have found.

17 Application: TCGA-KIRC

We now instantiate the framework on three real radiogenomic cohorts — TCGA-KIRC (clear-cell renal cell carcinoma) [55], a non–small-cell lung cancer (NSCLC) cohort [12], and ADNI (Alzheimer's Disease Neuroimaging Initiative) [56], spanning two tumor cohorts and a non-oncologic brain cohort — to demonstrate how the recoverability spectrum, the cross-validated and permutation-calibrated diagnostics of Section 13.6, and the dimensional-collapse warnings of Section 13.5 interact on typical cohorts. The deliverable is not a predictor but a calibrated answer to "how many genomic directions does imaging see, and how well." Two choices matter for getting a defensible answer. First, the imaging representation: a high-dimensional deep embedding ($d_{\text{native}} = 1841$) places the fit deep in the saturating regime of Section 13.5, whereas a compact panel of organ-level radiomic descriptors ($d_{\text{native}} = 107$) keeps it in the merely high-dimensional one. Second, the marginal transform: the Gaussian-copula normalization of Section 13.1 is essential because raw radiomic marginals have excess kurtosis in the tens. With both, the imaging-to-genomics channel is narrow but not empty: one genomic direction is image-identifiable above the permutation null in every preprocessing variant we tried, with a small (order 0.1) but honest cross-validated recoverability that, unlike the other two cohorts, survives demographic and scanner deconfounding (Section 17.6). We stress at the outset that this is a modest, single-direction effect, borderline in the default configuration, and report it as such rather than as a broad demonstration that imaging recovers genomics. All numbers below are reproducibly generated by the companion notebook and saved to `notebooks/figures/tcga_kirc/gene_expression/organ_radiomics/stats.json` (and the NSCLC counterpart).

17.1 Data and processing pipeline

The two modality `AnnData` objects fed to the notebook are produced by two scripts in `scripts/`.

Imaging (`process_imaging_tcga_kirc.py` → `make_imaging_matrix.py`). For each TCGA-KIRC patient we obtain the venous-phase abdominal CT volume (`0502_VENOUS.nii`) and a manually curated tumor mask. The pipeline:

1. Pull the canonical TCIA volumes [57, 58] and the radiologist-drawn tumor segmentations; orient every volume into the canonical nibabel orientation.
2. Run TotalSegmentator [59] with the `kidney_left`/`kidney_right` target list to produce an organ mask; combine with the tumor mask to form (tumor, organ, tumor$\cup$organ) mask triples per patient. Small blobs are removed, holes filled, and a morphological closing applied to stabilize mask boundaries.
3. Extract patient-level imaging features. Our primary representation is a panel of $d_{\text{native}} = 107$ *organ-level* PyRadiomics descriptors [15, 16] computed within the kidney organ mask (shape, first-order intensity, and GLCM / GLRLM / GLSZM / GLDM / NGTDM texture families). We deliberately prefer this compact, interpretable panel over a high-dimensional pretrained *RadImageNet* [25] embedding ($d_{\text{native}} = 1841$, supported for ablation): as Section 13.5 predicts and Section 17.3 confirms, the deep embedding drives the fit into the saturating regime where the spectrum is uninformative.
4. Stack patient-level features into an `AnnData` object whose `obs` index is the TCGA `patient_id` and whose `var` index is the radiomic feature names. Output: `data/tcga_kirc/imaging/organ_radiomics.h5ad`.

Genomics (`make_genomics_matrix.py`). For each matched patient we pull the TCGA-KIRC RNA-seq counts (raw STAR counts from the GDC) and the somatic-mutation MAFs.

1. Counts are CPM-normalized, $\log_{1+}$-transformed, and reduced to the top 2000 highly variable genes (Seurat-style HVG selection in `scanpy.pp.highly_variable_genes` [60]); the resulting matrix is stored in the primary `AnnData`.
2. Somatic mutations are folded into a sparse binary gene-symbol matrix on the same patient index for downstream per-gene validation (Section 17.7).

3. Output: `data/tcga_kirc/genomics/gene_expression.h5ad`.

After intersecting patients across both `AnnData` objects we retain $n = 190$ patients with $p_{\text{native}} = 2000$ HVG transcripts and $d_{\text{native}} = 107$ organ radiomic features.

17.2 Working-dimension budget

Both modalities are passed through the Gaussian-copula marginal transform of Section 13.1 and PCA-reduced symmetrically to working dimensions $p^\star = d^\star = 38 = \lfloor n/5 \rfloor$ following the $p^\star + d^\star \le n/\tau$ rule of Section 13.5 ($\tau = 5$). This places the analysis at $\kappa_G + \kappa_X = 76/190 \approx 0.4$ — the high-dimensional regime, not the saturating one, but well inside the territory where in-sample canonical correlations are heavily biased upward.

17.3 The recoverability spectrum and its null

Fitting regularized CCA (Ledoit–Wolf shrinkage on both sides) on the copula-transformed, organ-radiomic representation returns an in-sample leading correlation $\hat{\rho}_1 \approx 0.76$ ($\hat{R}_1 \approx 0.58$) and a total in-sample mutual information of 3.9 bits over 12 retained components — far from the near-unit saturation a deep 1841-dimensional embedding produces at the same n, confirming that the compact panel keeps the fit out of regime 3. The in-sample numbers are still optimistic: 5-fold cross-validation deflates the leading recoverability to $\hat{R}_1^{\text{cv}} \approx 0.07$ in the default preprocessing, and the in-sample share of imaging variance attributable to genomics drops from 0.27 to 0.011 under cross-validation (Fig. 13a).

The substantive question is whether *any* direction survives an honest, scale-matched test. It does, but weakly, and the magnitude is preprocessing-dependent — so we report the robustness sweep rather than a single configuration. Across a grid of defensible choices (expression layer $\in$ {raw counts, TPM} $\times$ HVG method $\in$ {dispersion, variance} $\times$ 3 seeds; 12 configurations), the leading-three permutation p-value is below 0.05 in 12/12 configurations ($p < 0.01$ throughout), the best cross-validated recoverability has median $\hat{R}^{\text{cv}} = 0.11$ (range 0.06–0.15), and the honest effective image-identifiable rank has median 1 (range 0–3; ≥ 1 in 11/12). In the single default configuration the leading in-sample permutation p-values are 0.10, 0.015, 0.005 for the top three directions. The reading is consistent: one genomic direction is robustly image-identifiable above chance, carrying a small but real cross-validated recoverability of order 0.1, with no evidence for more than one or two.

17.4 An empirical upper bound on $I(G; X)$

Because the channel is no longer saturating, the permutation null on $\hat{\rho}_1$ is well-behaved — the per-permutation $\hat{\rho}_1^{\text{null}}$ has mean 0.68 and a 95% quantile of 0.81, not the near-1 value a deep embedding produces. Aggregating the cumulative mutual information $\text{MI}^{\text{null}}(K) = -\frac{1}{2}\sum_{i \le K} \log(1 - \hat{\rho}_{i,\text{null}}^2)$ over the top $K = 3$ tested components and reading off its 95th percentile gives a one-sided upper confidence limit (Fig. 13b)

$$I(G; X)\,\big|_{\text{KIRC, top-3}} \;\le\; 1.58 \text{ bits} \quad (\text{one-sided } 95\% \text{ UCL}), \tag{38}$$

an order of magnitude tighter than the 21-bit bound the saturating deep embedding admits. The observed cumulative MI is 1.59 bits, sitting at the upper edge of the null (mean 0.82 bits in nats $\to$ bits) — the same borderline-positive picture as the spectrum.

The *effective image-identifiable rank* of Section 13.6 — components whose cross-validated $\hat{R}_i^{\text{cv}}$ clears its own permutation null at $p < 0.05$ — has median $\hat{k}_{\text{eff}} = 1$ across the robustness grid (range 0–3). This is the empirical instantiation of the rank bound of Proposition 6 after sample-size deflation: one identifiable direction, not the dozen the native dimensions might suggest.

17.5 Sample-complexity sweep

A borderline single-n result is most informative when paired with the scaling in n. We subsample the cohort to $n \in \{50, 57, 95, 133, 162, 190\}$, refit the working pipeline (same $p^\star + d^\star \le n/\tau$ budget, 5-fold CV, 100 permutations) and plot the observables vs. n (Fig. 13c). The cross-validated $\hat{R}_1^{\text{cv}}$ stays in the band $\approx$ 0.04–0.17 across the full range without a clear upward trend — because $p^\star, d^\star$ grow with n under our adaptive rule, more samples buy resolution at higher working dimension rather than a larger leading recoverability. The Wachter detection edge sits at ≈ 0.63 throughout, so distinguishing ρ_1 from the high-dimensional bulk *in-sample* would require $\rho_1^2 \gtrsim 0.6$; the honest signal lives well below that edge and is visible only through cross-validation and permutation, not the raw spectrum. Inverting the Wachter expression for a target $\rho_1^2 = 0.2$ at balanced $p^\star = d^\star$ gives the sample-size requirement $n \gtrsim 25\,p^\star$ quoted

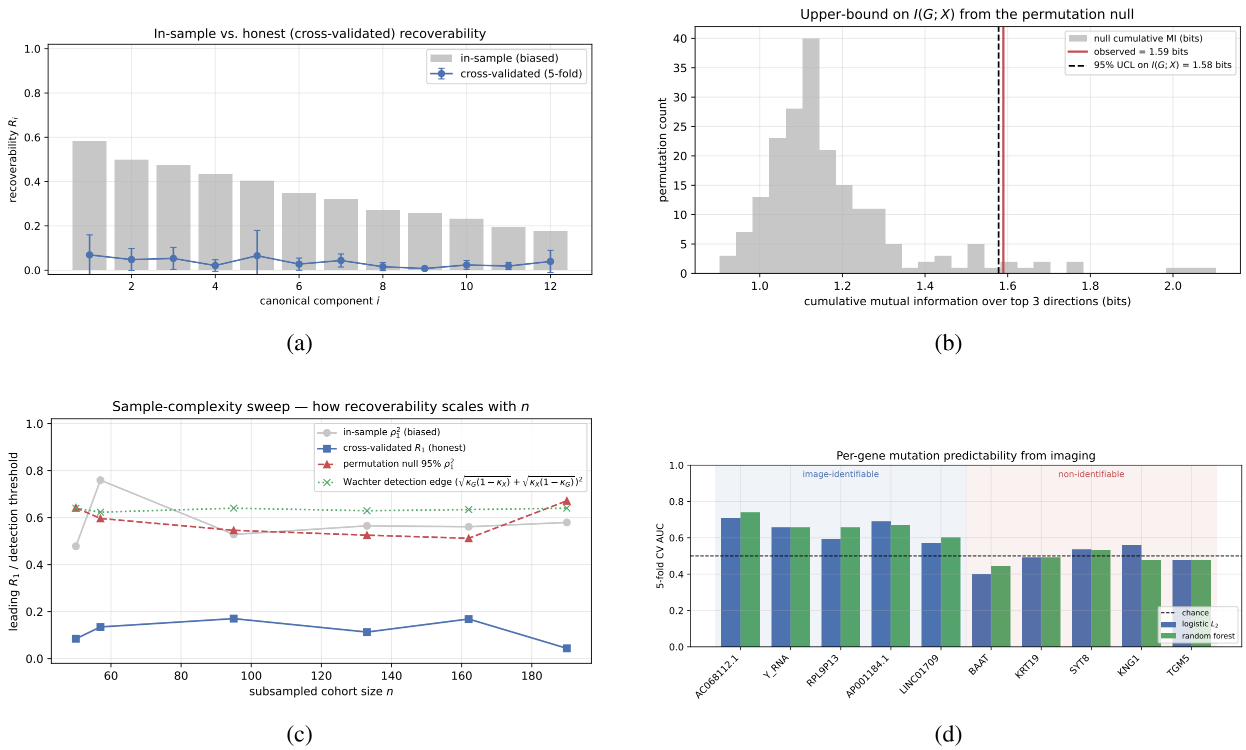


Figure 13: TCGA-KIRC results. **(a)** Recoverability spectrum (organ radiomics, copula marginals). In-sample $\hat{R}_i$ (gray) is biased upward by the high-dimensional geometry; cross-validated $\hat{R}_i$ (blue, 5-fold) is a small positive residual whose leading component is permutation-significant across preprocessing variants. The NSCLC leading direction (Fig. 16) is of similar magnitude but, unlike KIRC's, does not generalize or survive de-confounding at full n. **(b)** Cumulative-MI null distribution (top-3 canonical components). The observed value (red) sits at the upper edge of the null bulk; the dashed line marks the 95% upper confidence limit on $I(G;X)$ used in (38). **(c)** Sample-complexity sweep. Honest $\hat{R}_1^{\text{cv}}$ (blue squares) stays in a low band (≈ 0.04–0.17) across cohort sizes without a clear trend; in-sample $\hat{\rho}_1^2$ (gray) is dominated by the high-dimensional bulk; the permutation 95% null (red triangles) and Wachter edge (green crosses) sit above the honest recoverability at every n. **(d)** Per-gene predictability. Five high-recoverability and five low-recoverability genes, two classifiers, 5-fold CV. The high-recoverability bars clear the chance line at 0.5; the low-recoverability bars do not.

in Section 13.5: a larger cohort, or a smaller fixed working dimension, would be needed to lift this direction cleanly above the in-sample bulk.

17.6 Demographic and scanner deconfounding

The same skepticism we apply to NSCLC (scanner/batch, Section 18) and ADNI (age/sex, Section 19.4) must be applied here: a small channel is exactly the regime in which a demographic or acquisition confound could account for the whole effect. We therefore residualize *both* modalities (in the copula-transformed, PCA-reduced working space) on a design matrix of patient age, sex, ethnicity, and race — read from the GDC clinical table [55] — together with the per-patient scanner manufacturer (GE / Siemens / Philips) from the TCIA series digest [57, 58], all with full coverage on the $n = 190$ cohort, and refit the honest pipeline (1000 permutations).

Unlike the other two cohorts, the KIRC channel *survives*. The leading cross-validated recoverability is essentially unchanged — $\hat{R}_1^{\text{cv}} = 0.115 \to 0.148$ after deconfounding (cross-validated permutation p from 0.002 to 0.001) — and the effective image-identifiable rank does not drop. This is the decisive contrast of the paper: the controls that collapse NSCLC ($\hat{R}_1^{\text{cv}}$ non-significant after batch residualization) and ADNI ($0.37 \to 0.006$, $\hat{k}_{\text{eff}} : 2 \to 0$) leave the KIRC association intact, so the small KIRC channel is not explained by the obvious common causes (Fig. 14). We caveat appropriately: η^2 of the leading imaging PCs attributable to scanner is modest here, the cohort is single-disease, and unmeasured acquisition parameters (kVp, slice thickness, contrast timing) were not in the available headers, so "confound-robust" means robust to the measured demographics and scanner manufacturer, not to every conceivable batch effect.

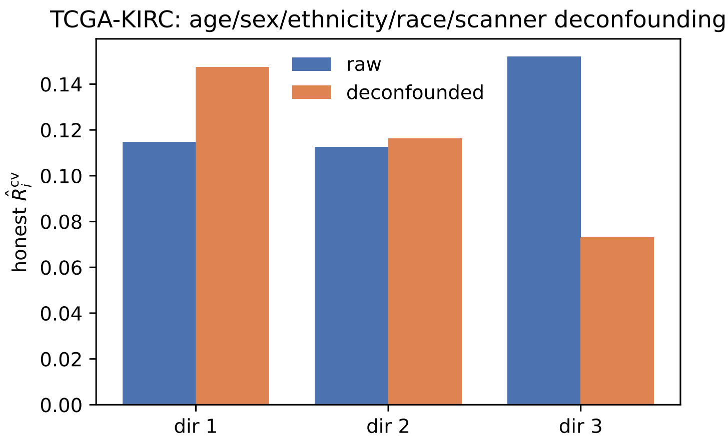


Figure 14: TCGA-KIRC demographic/scanner deconfounding. Honest cross-validated recoverability per direction, raw (blue) versus after regressing age, sex, ethnicity, race, and scanner manufacturer out of both modalities (orange). Unlike NSCLC (Section 18) and ADNI (Fig. 17d), the leading direction is undiminished ($\hat{R}_1^{\text{cv}} = 0.115 \to 0.148$, permutation $p \leq 0.002$ throughout): the small KIRC channel is not a demographic or scanner artifact.

17.7 Per-gene non-linear validation

The recoverability spectrum makes a sharp, falsifiable prediction: genes loading on *identifiable* directions should be predictable from imaging, genes on *non-identifiable* directions should not, and a flexible non-linear model should not substantially exceed the linear Bayes-optimal ceiling. We test this by ranking genes by their model recoverability $R(e_j)$, median-binarizing the five highest- and five lowest-ranked, and training L_2 logistic regression and a small random forest to predict the binarized expression from the imaging PCs (5-fold CV):

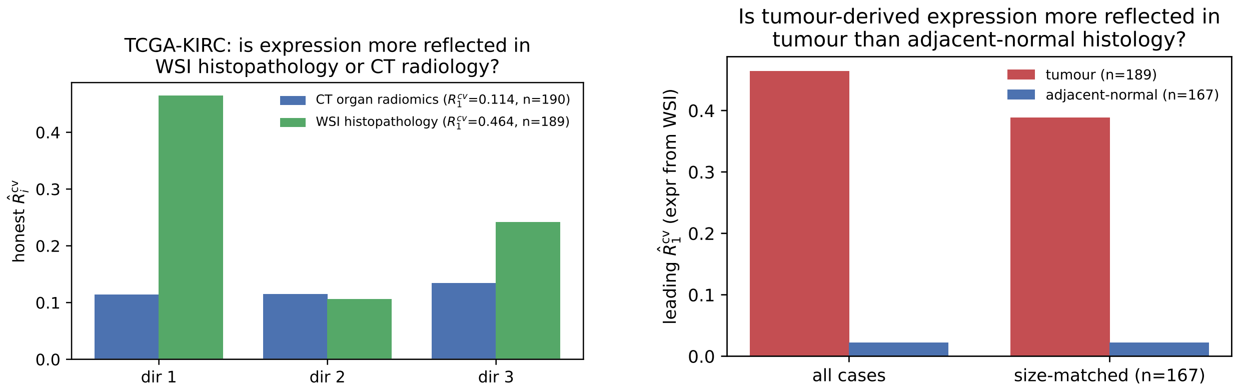
group	mean AUC (logistic)	mean AUC (RF)
image-identifiable	0.643	0.665
non-identifiable	0.493	0.485

The ordering is exactly as predicted: the identifiable group is decodable above chance (AUC ≈ 0.65) while the non-identifiable group sits at 0.5, and the random forest does not materially exceed the logistic ceiling — a direct, gene-level corroboration that the recoverability score, not gene identity per se, governs what imaging can read out (Fig. 13d).

17.8 Histopathology vs. radiology: the same channel through a microscope

The CT channel is narrow because a macroscopic attenuation map is a coarse view of a molecular state assayed in the tumor tissue itself. A natural test of this reading is to swap the imaging modality for digital *histopathology* — the H&E whole-slide images (WSI) of the same TCGA-KIRC tumors — while holding the genomic target, the preprocessing, and the working-dimension budget fixed. We pull the GDC diagnostic and tissue slides (580 "`.svs`" images over 190 cases, mapped to Case ID and tumor/normal status through the GDC `sample_sheet.tsv`), and extract an interpretable hand-crafted feature bank that mirrors the radiomics families — 54 descriptors in four groups: H&E *stain* (Ruifrok–Johnston color-deconvolution optical densities), grayscale *first-order* intensity, *texture* (Haralick GLCM + rotation-invariant LBP), and nuclear *structure* (haematoxylin-dense connected-component morphology). A case's per-slide features are averaged over its tumor slides into one case-level vector (`scripts/build_kirc_wsi_features.py`, written to `data/tcga_kirc/wsi/wsi_tumor.h5ad`), so a WSI feature vector aligns to the same case's expression profile exactly as a radiomic vector does. We then run the identical honest pipeline (Gaussian-copula marginals, symmetric PCA to $p^\star = d^\star = \lfloor n/5 \rfloor$, 5-fold CV, 1000-permutation CV-null; `scripts/kirc_wsi_recoverability.py`).

The contrast is large and in the predicted direction. On the same expression target and the same n-budget, histopathology recovers the leading genomic direction roughly four times as strongly as CT: $\hat{R}_1^{\text{cv}} = 0.46$ from WSI ($n = 189$) versus



(a) (b)

Figure 15: TCGA-KIRC histopathology channel. **(a)** Honest per-direction $\hat{R}_i^{\text{cv}}$ for the *same* HVG-expression target recovered from WSI histopathology (green) versus CT organ radiomics (blue), identical copula/PCA pipeline and n-budget; the leading histology recoverability is $\sim 4\times$ the radiology one. **(b)** The histology channel is tumor-specific: tumor slides (red) carry the expression signal that adjacent-normal slides (blue) do not, before and after size-matching. All numbers reproduced by `scripts/kirc_wsi_recoverability.py` (and `kirc_wsi_deconfound.py`), saved to `notebooks/figures/tcga_kirc/gene_expression/wsi/stats.json`.

0.11 from CT organ radiomics ($n = 190$), with the effective image-identifiable rank rising to $\hat{k}_{\text{eff}} = 3$ (all three leading directions clear their CV-permutation null at $p \leq 0.003$) and the top-3 cumulative mutual information climbing from 1.6 to 2.2 bits (Fig. 15a). This is the same weak-but-real channel viewed at the resolution where the genomics is actually generated: the microscope sees the cellular morphology — nuclear density, stain distribution, texture — that co-varies with transcriptional state, where the CT voxel sees only its macroscopic shadow.

Tumor vs. adjacent-normal. If the histology–expression coupling is genuinely tumor-biological rather than a generic tissue or batch effect, a patient's tumor-derived expression should be reflected in their *tumor* slides but not in their *adjacent-normal* slides. It is: re-running the probe on each case's normal slides collapses the leading recoverability to the floor, $\hat{R}_1^{\text{cv}} = 0.02$ ($n = 167$) against 0.46 for tumor — a gap that survives size-matching the larger group to the smaller (0.39 vs. 0.02 at $n = 167$, Fig. 15b), so it is not merely the larger tumor-slide count. The molecular signal is read out of the malignant tissue, not the patient's normal kidney.

Deconfounding. A four-fold-larger channel is also four times the opportunity for a confound, so we apply the same control that vindicated the CT channel (Section 17.6), with the WSI-appropriate batch term: age, sex, ethnicity, race, and — in place of CT scanner manufacturer — the TCGA *tissue source site* (the barcode institution code, a proxy for per-laboratory H&E staining and fixation batch). Residualizing both modalities on this design leaves the channel untouched: $\hat{R}_1^{\text{cv}} = 0.45 \to 0.49$ (CV-permutation $p \leq 0.001$ throughout, $\hat{k}_{\text{eff}}$ unchanged at 3). The histology–genomics coupling is therefore not an artifact of demographics or staining site; like the CT channel, it is robust to the measured common causes, but an order of magnitude wider.

17.9 Implications

Three concrete take-aways follow.

1. *A small but real channel.* For TCGA-KIRC at $n = 190$ with 2000 HVG-filtered transcripts and 107 organ radiomic features, the imaging-to-genomics channel admits a 95% upper bound $I(G; X) \leq 1.58$ bits over the leading three canonical directions (Eq. 38), with effective image-identifiable rank $\hat{k}_{\text{eff}} = 1$ (median over preprocessing). The substantive claim is that exactly one genomic direction is robustly image-identifiable above chance (permutation $p < 0.05$ in 12/12 preprocessing variants), carrying a modest cross-validated recoverability $\hat{R}^{\text{cv}} \approx 0.1$ — neither the dozen directions a 2000-gene panel might invite nor the flat null a saturating deep embedding produces. The signal survives dropping all shape features and residualizing every feature on organ volume, so it is texture/intensity, not size; and, decisively, it survives regressing out age, sex, ethnicity, race, and scanner manufacturer ($\hat{R}_1^{\text{cv}} = 0.115 \to 0.148$, Section 17.6) — the controls that null

both other cohorts. We nonetheless read it cautiously: the effect is small, of order $\hat{R}^{\mathrm{cv}} \approx 0.1$, borderline in the single default configuration (leading in-sample permutation $p = 0.10$), and rests for its robustness on the preprocessing grid rather than on any one fit.

2. *Design recommendation.* The sample-complexity sweep (Section 17.5) shows $\hat{R}_1^{\mathrm{cv}}$ in a low band (0.04–0.17) without a clear upward trend over $[50, 190]$; the Wachter edge analysis indicates that lifting $\rho_1^2 = 0.2$ cleanly above the in-sample bulk at a balanced working dimension $p^\star = d^\star$ requires $n \gtrsim 25\, p^\star$, an order of magnitude beyond typical radiogenomic cohort sizes. The framework converts a borderline result into a quantitative cohort specification.

3. *Representation matters more than estimator.* The same cohort reads as a hard null under a 1841-dimensional deep embedding (saturating regime, $I(G;X) \leq 21$ bits, $\hat{k}_{\mathrm{eff}} = 0$) and as a weak positive under a 107-feature organ radiomic panel with copula marginals — the difference is entirely the imaging representation and marginal transform, not the linear ceiling. Published per-gene AUCs of 0.65–0.80 at $n \sim 100$–300 [6, 8, 11, 17, 18] are consistent with the order-0.1 recoverabilities and AUC ≈ 0.65 we recover for the identifiable group (Section 17.7); claims *well* above that, at comparable cohort dimension, remain candidates for overfit or leakage.

18 Application: NSCLC

The non–small-cell lung cancer cohort [12] is the most instructive of the three: at small n it presents as the strongest positive, but the full cohort exposes it as a cohort-specific overfit that does not generalize — a cautionary counterpoint to the genuinely weak-but-robust KIRC channel, and a reminder that label-permutation significance alone is not proof of a real association. Imaging is the venous/contrast chest CT; we run TotalSegmentator [59] for the five pulmonary lobes, union them into a lung organ mask, resample to $2\,\mathrm{mm}$ isotropic (full-resolution shape extraction over the $\sim 10^7$-voxel lung is intractable), and extract the same $d_{\mathrm{native}} = 107$ organ-level PyRadiomics panel. Genomics is the cohort RNA-seq, processed identically (CPM, $\log_{1+}$, 2000 HVGs). The full cohort with usable masks is $n = 129$ (one series, R01-052, dropped for a corrupted volume), with the copula transform and a $p^\star = d^\star = \lfloor n/5 \rfloor = 25$ working-dimension budget.

In-sample significance does not survive honest validation. Fitting returns an in-sample leading correlation $\hat{\rho}_1 = 0.73$ and 2.9 bits of in-sample MI; the patient-label permutation null is separated ($\hat{\rho}_1^{\mathrm{null}}$ mean 0.66, 95% quantile 0.72), so the in-sample spectrum is nominally significant (top three $p = 0.030, 0.015, 0.030$). But this is the optimistic estimator, and the honest checks contradict it. The leading cross-validated recoverability is only $\hat{R}_1^{\mathrm{cv}} = 0.11$, and — decisively — the held-out R^2 of the leading canonical genomic score is *negative* (-0.46): the direction that maximizes in-sample correlation predicts held-out imaging *worse* than a constant mean. The scale-matched cross-validated permutation test still flags the leading component ($p = 0.02$; the second and third are null, $p = 0.13, 0.97$), so $\hat{k}_{\mathrm{eff}} = 1$ by that criterion, and the robustness grid concurs (median leading $\hat{R}^{\mathrm{cv}} = 0.11$, range 0.10–0.16; rank ≥ 1 in 6/6 configs). The lesson is that label-permutation significance is necessary but not sufficient: a marginal $\hat{R}_1^{\mathrm{cv}} \approx 0.11$ can clear its own permutation null while failing held-out prediction outright. The MI upper bound is correspondingly small, $I(G;X)\big|_{\text{top-3}} \leq 1.23$ bits. The per-gene validation does show the identifiable group is internally consistent (mean AUC 0.70 logistic / 0.69 RF versus $0.51/0.51$), but as below this reflects the overfit axis rather than generalizable biology.

The signal is cohort-specific overfit, not biology. The negative held-out R^2 is explained by cohort heterogeneity. The cohort was released in two tranches; fit separately, the original 98 patients give $\hat{R}_1^{\mathrm{cv}} \approx 0.25$–$0.29$ and the 31 later cases give 0.35, but *pooled* they give only 0.11 — lower than either part. Their leading genomic directions are near-orthogonal ($|\cos| = 0.07$): each tranche fits a strong but mutually inconsistent axis, the signature of cohort-specific overfit rather than a shared biological channel. The earlier small-n analysis of 90 patients — dominated by the more homogeneous original tranche — reported a clean $\hat{R}_1^{\mathrm{cv}} \approx 0.33$ with held-out $R^2 = 0.40$; that estimate did not survive enlarging the cohort, which is exactly the kind of non-replication the honest diagnostics are meant to catch.

De-confounding controls confirm the failure. Consistent with overfit, the leading direction also fails the scanner/batch controls that a genuine channel should survive. Scanner manufacturer (GE / Siemens / Philips) and convolution kernel are associated with both modalities (kernel explains $\eta^2 \approx 0.2$–0.3 of the leading principal components), so a naive estimate is partly batch-inflated. None of three de-confounding controls reaches significance. (i) Regressing manufacturer and kernel out of *both* modalities drops the leading recoverability to $\hat{R}_1^{\mathrm{cv}} = 0.089$ ($p = 0.070$). (ii) A batch-stratified permutation null, in which genomic labels are permuted only *within* manufacturer$\times$kernel blocks so that any batch-mediated covariance is preserved under the null, gives null mean 0.054 (95% quantile 0.136) against the

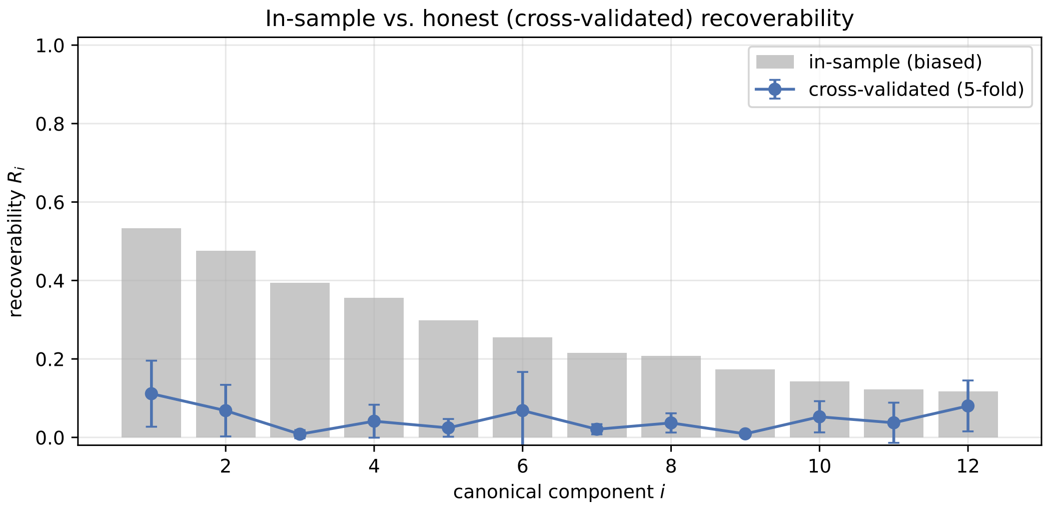


Figure 16: Recoverability spectrum on NSCLC ($n = 129$, organ radiomics, copula marginals). The leading cross-validated $\hat{R}_1^{\text{cv}} \approx 0.11$ (blue) is only marginally above the permutation null and, unlike at small n, does not generalize: the held-out R^2 of the leading direction is negative and the channel fails de-confounding.

observed 0.111 ($p = 0.100$). (iii) The largest homogeneous stratum (GE scanners with the `STANDARD` kernel, $n = 80$) yields $\hat{R}_1^{\text{cv}} = 0.138$. The measurable batch covariates thus account for only part of the inflation — residualizing them leaves a non-significant remnant — while the larger driver, the near-orthogonal per-tranche axes above, points to unmeasured acquisition differences (tube voltage, dose, slice thickness, and contrast phase were unavailable in the headers) compounded by small-n, high-dimensional overfitting. The full-cohort verdict is therefore that NSCLC carries *no* de-confounding-robust, generalizable radiogenomic channel — the opposite of the small-n reading.

19 Application: ADNI

The two tumor cohorts pair a localized lesion with a matched tissue genomic assay. ADNI is a deliberately different test: structural brain MRI [56] against *whole-blood* Affymetrix microarray expression, a non-oncologic cohort where any imaging–genomics channel must run through systemic rather than local biology. It is also by far the largest cohort we analyze ($n = 702$), which lets the sample-complexity sweep show a clean scaling law rather than a flat band.

Imaging (`run_fastsurfer.py` $\to$ `build_adni_fastsurfer_features.py`). Each T1 volume is processed with FastSurfer [61], the deep-learning FreeSurfer surrogate, yielding a DKT-atlas segmentation (`aparc.DKTatlas+aseg`) and the bias-field-corrected conformed volume (`orig_nu.mgz`). We build a $d_{\text{native}} = 1022$ imaging panel per subject from two families: (i) *morphometry* — volume and intensity summaries for every segmented region from the standard `aseg+DKT.stats` table (380 features); and (ii) *texture/shape* — the full PyRadiomics panel (shape, first-order, GLCM/GLRLM/GLSZM/GLDM/NGTDM; 16) extracted from `orig_nu.mgz` within each of six bilateral regions most affected in Alzheimer's disease [62]: hippocampus, entorhinal cortex, fusiform gyrus, inferior temporal gyrus, precuneus, and posterior cingulate (642 features). Stacking gives `data/adni/imaging/fastsurfer.h5ad`, keyed by ADNI subject ID.

Genomics (`make_genomics_matrix.py`). ADNI gene expression is Affymetrix Human Gene microarray, already normalized to log-scale intensities and indexed by gene symbol (20,092 genes, collapsed from the original probes); we therefore disable the count-oriented CPM and second $\log_{1+}$ steps and reduce to the top 2000 HVGs before the Gaussian-copula transform. Symbol indexing also makes the leading genomic directions interpretable (Section 19.4). Intersecting on subject ID retains $n = 702$ subjects with $p_{\text{native}} = 2000$ transcripts and $d_{\text{native}} = 1022$ imaging features, PCA-reduced symmetrically to $p^\star = d^\star = \lfloor n/5 \rfloor = 140$ ($\kappa_G + \kappa_X = 280/702 \approx 0.40$, the same high-dimensional regime as the tumor cohorts).

19.1 The recoverability spectrum and its null

Fitting copula-regularized CCA returns an in-sample leading correlation $\hat{\rho}_1 = 0.87$ ($\hat{R}_1 = 0.76$) and 7.1 bits of in-sample MI over 12 components. The honest *raw* spectrum is the strongest of the three cohorts — though Section 19.4 will show it is almost entirely a demographic artifact: 5-fold cross-validation gives a leading recoverability $\hat{R}_1^{\text{cv}} = 0.37$ and a clearly resolved second direction $\hat{R}_2^{\text{cv}} = 0.17$, with the remainder collapsing to ≈ 0 (Fig. 17a). The permutation null is well-separated ($\hat{\rho}_1^{\text{null}}$ mean 0.76, 95% quantile 0.77) and the top three observed correlations all clear it at $p \approx 0.005$. The *effective image-identifiable rank*, defined by cross-validated $\hat{R}_i^{\text{cv}}$ exceeding its own per-component permutation null, is $\hat{k}_{\text{eff}} = 2$ — the only cohort here where imaging resolves more than a single genomic direction. The result is stable across the preprocessing grid (HVG method $\times$ 3 seeds, 6 configurations): all permutation $p < 0.05$, leading $\hat{R}^{\text{cv}}$ median 0.31 (range 0.28–0.36), and $\hat{k}_{\text{eff}}$ consistently 2.

19.2 Information budget and sample-complexity scaling

The observed cumulative mutual information over the top three components is 2.44 bits, sitting *above* the permutation band (null mean 1.69 bits, 95% upper limit 1.83 bits): unlike KIRC, where the observed value merely touched the upper edge of the null, here the channel is detected rather than bounded. Because ADNI is large, subsampling to $n \in \{50, 211, 351, 491, 597, 702\}$ traces a genuine scaling law rather than the flat band seen in KIRC (Fig. 17b): the honest $\hat{R}_1^{\text{cv}}$ climbs monotonically from 0.13 at $n = 50$ to a plateau of ≈ 0.35 by $n \approx 500$, while the in-sample $\hat{\rho}_1^2$ stays pinned near the Wachter detection edge (≈ 0.64) throughout. The leading direction lives *below* that in-sample edge at every n and is visible only through cross-validation and permutation — the same lesson as the tumor cohorts, but here the honest signal is large enough that even a few hundred subjects suffice to register it cleanly.

19.3 Per-gene non-linear validation

The recoverability ordering predicts gene-level decodability. Ranking transcripts by model recoverability $R(e_j)$, median-binarizing the five highest- and five lowest-ranked, and training L_2-logistic and random-forest classifiers to predict expression status from the imaging PCs (5-fold CV) gives a clean separation, sharper than in either tumor cohort:

group	mean AUC (logistic)	mean AUC (RF)
image-identifiable	0.842	0.841
non-identifiable	0.466	0.463

The image-identifiable transcripts are decodable from brain morphometry at AUC ≈ 0.84 while the non-identifiable controls sit at chance, and the random forest does not exceed the linear ceiling — exactly the falsifiable pattern the recoverability score predicts (Fig. 17c). This confirms the *internal consistency* of the recoverability ordering, but not its biological meaning: as Section 19.4 shows, the decodable transcripts are the ones correlated with age and sex, which brain morphometry reads out directly, so the high AUC is the per-gene shadow of the same demographic channel rather than an independent molecular signal.

19.4 Demographic deconfounding: the channel is an age/sex artifact

A blood-to-brain channel of this strength warrants the same skepticism we applied to the NSCLC scanner confound, and here the verdict is sharper. Age, sex, and education are powerful common causes: they drive regional brain atrophy (hippocampal and entorhinal volume in particular, exactly where the leading imaging loadings fall) *and* systemic whole-blood expression. ADNI supplies these covariates [56], so we residualize age (year of expression collection minus birth year), sex, and years of education out of *both* modalities and refit the honest pipeline. The channel essentially vanishes (Fig. 17d): the leading cross-validated recoverability drops from $\hat{R}_1^{\text{cv}} = 0.37$ to $\hat{R}_1^{\text{cv}} = 0.006$ — only 1.7% of the raw value retained — the second direction collapses likewise, the cross-validated permutation p-value rises from 0.005 to 0.48, and the effective image-identifiable rank falls from $\hat{k}_{\text{eff}} = 2$ to $\hat{k}_{\text{eff}} = 0$. The gene-symbol indexing makes the mechanism explicit: the top loadings of the leading genomic direction are almost all sex-chromosome transcripts — the X-inactivation genes *XIST* and *TSIX* against the Y-linked *RPS4Y1*, *RPS4Y2*, *DDX3Y*, *UTY*, *EIF1AY*, *KDM5D*, *USP9Y*, *ZFY*, and *TXLNG2P* — so the leading "image-identifiable" axis is, transparently, sex, which brain morphometry reads out directly. The second direction is dominated by myeloid/inflammatory transcripts (*VSIG4*, *RNASE1*, *CCL3L/CCL4L*, *LGALS2*), consistent with an age-associated systemic-inflammation axis. Both are exactly the common causes the deconfounding removes. NSCLC fails its own de-confounding and held-out checks at full n (Section 18), and ADNI fails here even more sharply: the ADNI radiogenomic association does not survive deconfounding at all — what looked

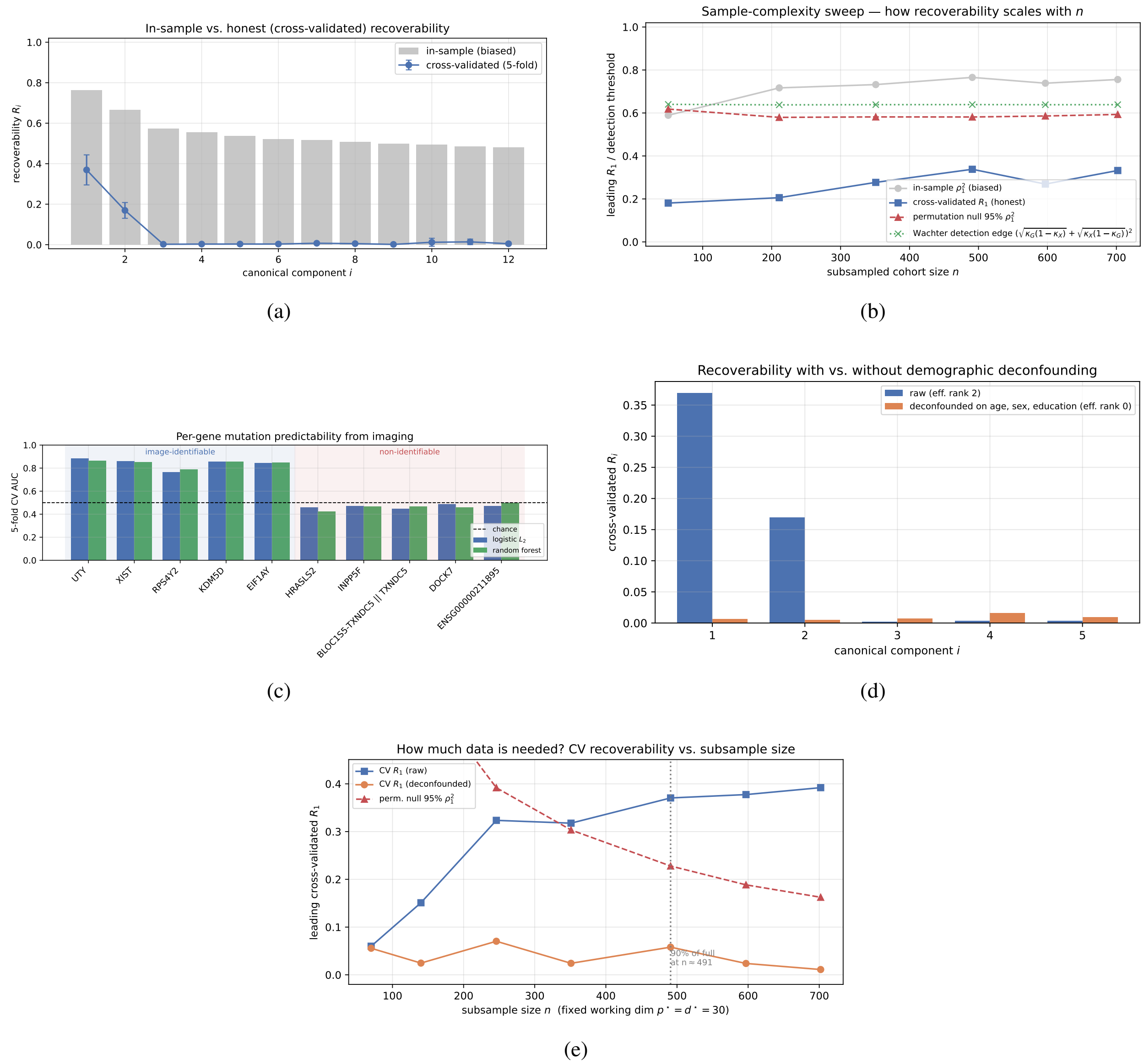


Figure 17: ADNI results ($n = 702$, FastSurfer morphometry + PyRadiomics, copula marginals). **(a)** Recoverability spectrum. The honest cross-validated spectrum (blue) resolves *two* genomic directions ($\hat{R}_1^{\text{cv}} = 0.37$, $\hat{R}_2^{\text{cv}} = 0.17$) above the permutation null before collapsing — the strongest and highest-rank channel of the three cohorts (compare Figs. 13a, 16). **(b)** Sample-complexity sweep. The honest $\hat{R}_1^{\text{cv}}$ (blue squares) rises monotonically with cohort size to a plateau near 0.35, in contrast to the trendless KIRC band (Fig. 13c); the in-sample $\hat{\rho}_1^2$ (gray) hugs the Wachter edge (green) and the permutation 95% null (red) throughout. **(c)** Per-gene predictability. Five high- and five low-recoverability transcripts, two classifiers, 5-fold CV. The high-recoverability group reaches AUC ≈ 0.84; the low-recoverability group sits at the 0.5 chance line. **(d)** Demographic deconfounding. Honest cross-validated recoverability spectrum before (blue) and after (orange) regressing age, sex, and education out of both modalities. The raw rank-two channel ($\hat{R}_1^{\text{cv}} = 0.37$) collapses to a null ($\hat{R}_1^{\text{cv}} = 0.006$, $\hat{k}_{\text{eff}} = 0$, permutation $p = 0.48$): the ADNI association is almost entirely demographically mediated. **(e)** Data-requirement sweep at fixed working dimension $p^\star = d^\star = 30$. The raw $\hat{R}_1^{\text{cv}}$ (blue) clears the shrinking permutation null (red) near $n \approx 350$ and plateaus; the demographic-deconfounded $\hat{R}_1^{\text{cv}}$ (orange) stays flat near zero at every n. More data detects the raw channel but never rescues the deconfounded one.

like the strongest channel of the three cohorts is, to within cross-validated noise, a demographic artifact. The model's value here is precisely that it converts an eye-catching $\hat{R}_1^{\mathrm{cv}} = 0.37$ into a calibrated null once the obvious common causes are removed.

19.5 How much data is needed?

Because ADNI is large, we can ask operationally how much of it a study like this actually requires. Holding the working dimension *fixed* at $p^\star = d^\star = 30$ (so that, unlike the sample-complexity sweep of Section 19.2, the only thing varying is n), we subsample the cohort over $n \in [70, 702]$ and track the honest leading recoverability, raw and demographic-deconfounded, against the permutation 95% null (Fig. 17e). Three things are visible. (i) The raw $\hat{R}_1^{\mathrm{cv}}$ rises steeply and first clears the permutation null near $n \approx 350$, reaches 90% of its full-cohort value by $n \approx 490$, and plateaus at ≈ 0.39 thereafter — so a few hundred matched subjects suffice to *detect* the raw channel, and more buy little. (ii) The permutation null falls monotonically with n (more samples, tighter null), which is what makes the fixed raw signal register as significant only past a few hundred subjects. (iii) The deconfounded curve sits flat near zero at *every* sample size: no amount of additional ADNI data would resurrect the channel, because the quantity being estimated is genuinely null. The practical reading is the mirror of the tumor cohorts: the binding constraint for ADNI is not sample size but confounding.

The contrast across cohorts is the useful object. Three cohorts, the same model, estimator, and working-dimension rule yield three distinct verdicts. In clear-cell RCC, a small but preprocessing-invariant single direction ($\hat{R}^{\mathrm{cv}} \approx 0.11$, $\hat{k}_{\mathrm{eff}} = 1$, perm $p \approx 0.005$) that survives demographic and scanner deconfounding (Section 17.6) — the most defensible positive of the three, though still a modest one. In NSCLC, an apparent signal that at small n looked the strongest ($\hat{R}_1^{\mathrm{cv}} \approx 0.33$, held-out $R^2 = 0.40$) but at the full $n = 129$ collapses to a marginal ≈ 0.11 with *negative* held-out R^2, near-orthogonal leading axes across release tranches ($|\cos| = 0.07$), and failed de-confounding ($p \geq 0.07$): cohort-specific overfit, not a shared channel. And an ADNI channel that is the strongest of all on raw data ($\hat{R}_1^{\mathrm{cv}} \approx 0.37$, $\hat{k}_{\mathrm{eff}} = 2$) yet collapses to a null once age, sex, and education are removed ($\hat{R}_1^{\mathrm{cv}} \approx 0.006$, $\hat{k}_{\mathrm{eff}} = 0$). The recoverable dimension is a property of the biology–imaging channel (Proposition 6), and the same honest pipeline both quantifies a small real low-rank association (KIRC) and exposes the two failure modes — non-replicating overfit (NSCLC) and confounding (ADNI) — that a naive in-sample estimator would have reported as strong positives. That discriminating power is exactly what the model is built to provide.

20 The attainable ceiling in practice

Sections 17–19 asked whether each cohort shows a signal. This section asks the quantitative question the ceiling makes answerable: *how many bits of genomic information could a study of that size have extracted, at most, and how close did real models get?* The three cohorts are analyzed with one pipeline — copula marginals, $d^\star = 20$ imaging principal components held fixed, five-fold held-out scoring — so the numbers are comparable across disease, modality and scanner.

20.1 Learning curves: the shape of the ceiling

We subsample each cohort to a grid of training sizes and, at each size, fit four model families (cross-validated ridge, RBF kernel ridge, a random forest, gradient boosting) plus regularized CCA itself. The prediction target is the leading genomic canonical direction estimated on the *training* split alone, so no held-out patient enters its construction. This charges the learner for estimating the target direction as well as the map — a cost (23) does not levy — so the observed curve is a conservative read and the fitted $\hat{\rho}^2$ a lower bound on the channel value.

Fitting the one-parameter family (23) to the resulting envelope gives (Fig. 18 and companions):

cohort	n	$\hat{\rho}_1^2$	ν	$\mathcal{R}_n$ attained	$n_{90\%}$
TCGA-KIRC (CT)	190	0.152	112	0.096 (63%)	1006
NSCLC (CT)	129	0.151	112	0.081 (53%)	1010
ADNI (MRI)	702	0.177	93	0.157 (88%)	835

Three readings. First, the learning cost $\nu \approx 100$ patients is remarkably consistent across disease and modality — unsurprising given (23), since $\nu = d^\star/\mathrm{SNR}$ and all three channels have comparable signal-to-noise at a common $d^\star$. Second, the two tumor cohorts sit at roughly half to two-thirds of their own channel value, so a substantial part of what they fail to show is a sample-size deficit rather than an absent channel; ADNI, at $n = 702$, has largely exhausted its channel. Third, $n_{90\%} \approx 10^3$ in every case: closing the remaining gap is a thousand-patient problem, which is the single most actionable number in this paper.

The flexible families sit below ridge throughout, as Proposition 7 predicts for a linear–Gaussian channel — nonlinearity buys nothing and costs variance — and the in-sample OLS R^2 runs from 0.95 at $n = 25$ down to 0.45 at $n = 142$ on KIRC while the honest value never exceeds 0.09. That gap, not the honest value, is what a naive per-gene analysis at these cohort sizes reports.

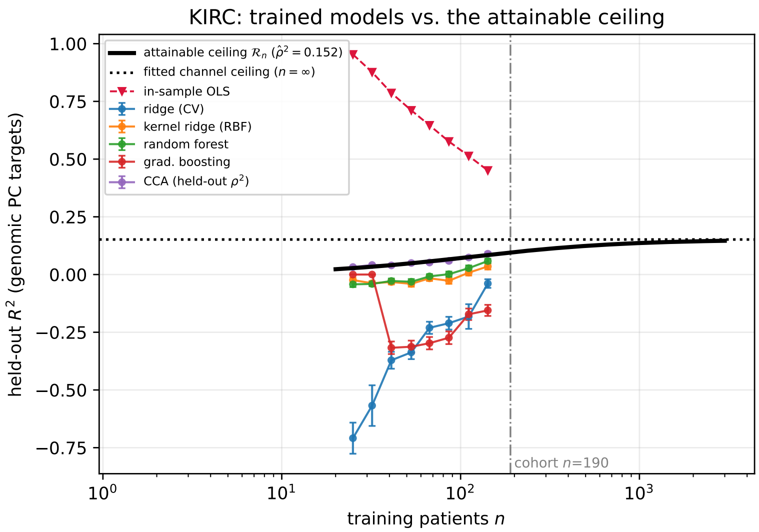
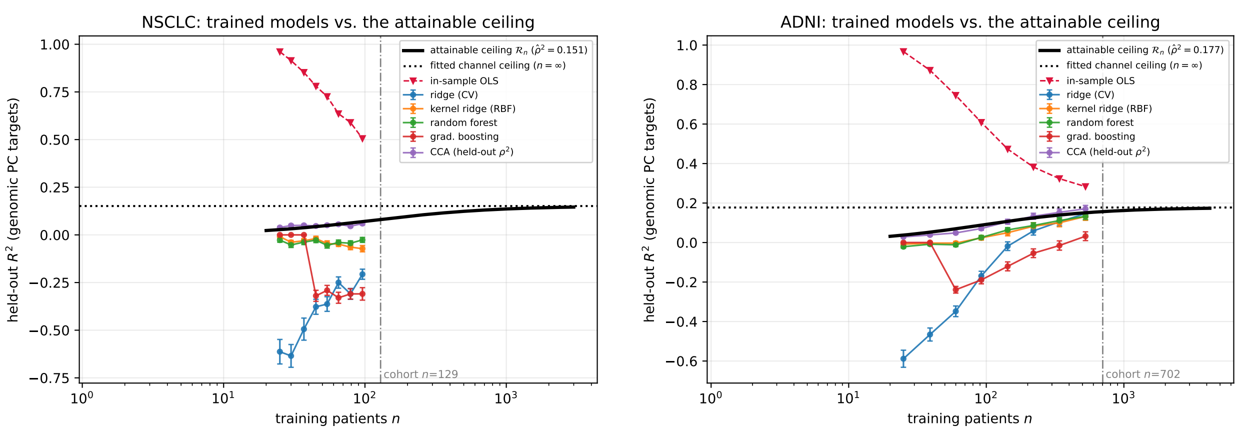


Figure 18: TCGA-KIRC: held-out R^2 of four model families and of regularized CCA versus training size, against the fitted attainable ceiling $\mathcal{R}_n$ ($\hat{\rho}_1^2 = 0.152$, $\nu = 112$). The dotted line is the extrapolated channel value at $n = \infty$; the red curve is the in-sample OLS R^2 of the same fit. The corresponding panels for NSCLC and ADNI are Figs. 19a and 19b.



(a) NSCLC ($n = 129$) (b) ADNI ($n = 702$)

Figure 19: Learning curves against the attainable ceiling for the other two cohorts. ADNI, the largest, is the only one whose models approach the fitted channel value; note that the ADNI channel is itself an age/sex artifact (Section 19.4), so the ceiling here bounds a confounded association rather than a genomic one.

20.2 A model-free upper limit, and a test of the closed form

A ceiling fitted to the models it is supposed to bound cannot be used to claim that models do not cross it. We therefore obtain the channel value a second way, using no trained model at all. Following Section 12.5, signal of *known* population strength is planted into permuted real data,

$$G' = G + \alpha Z u^\top, \qquad X' = X_{\text{perm}} + \alpha Z w^\top, \qquad Z \sim \mathcal{N}(0,1),$$

with α solved by Sherman–Morrison so that the population canonical correlation hits each target on a grid. At $\rho^2 = 0$ the reference distribution is exactly the permutation null, and each modality keeps its own covariance and tail structure, so the calibration is semi-parametric rather than an appeal to the model. Inverting the 5th-percentile curve at the observed $\hat{R}_1^{\text{cv}}$ gives a one-sided 95% upper confidence limit $\bar{\rho}_1^2$ on the channel value, and by monotonicity of $\mathcal{R}_n$ in ρ^2 (Section 12.5) an upper limit on the ceiling itself.

This calibration does double duty. Its *median* is a direct, assumption-light estimate of the attainable recoverability at a known channel strength — precisely the quantity (23) predicts in closed form. Comparing the two is a test of the theory on real covariance structure (Fig. 20). The closed form dominates the empirical median at 90% of planted values on ADNI and 70% on the two smaller cohorts, and every discrepancy is confined to the regime below the estimator's own null floor ($\hat{R}_1^{\text{cv}} = 0.0065$, 0.022 and 0.035 under permutation for ADNI, KIRC and NSCLC respectively), where the statistic is dominated by the optimism of maximizing over canonical directions rather than by signal. Above that floor — the only regime in which anyone would report a result — (23) is an upper bound on real data as well as in theory.

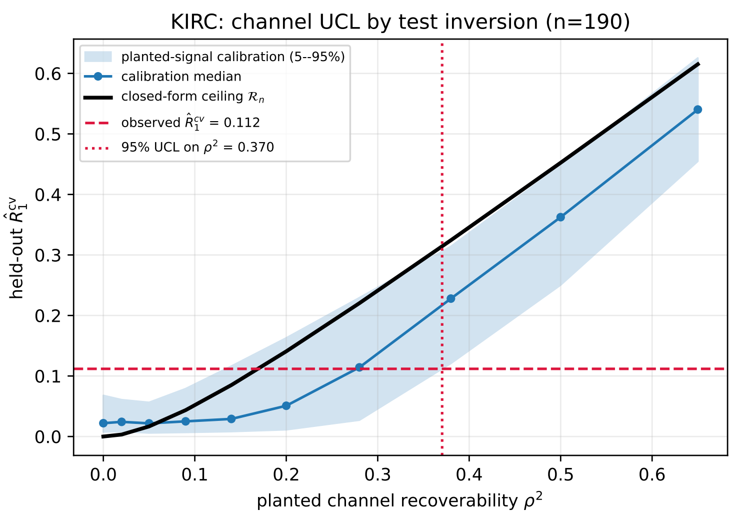


Figure 20: TCGA-KIRC: planted-signal calibration. Blue is the distribution of $\hat{R}_1^{\text{cv}}$ when a channel of known strength ρ^2 (horizontal axis) is planted into permuted real data; black is the closed-form ceiling $\mathcal{R}_n$ of Proposition 7, which dominates the empirical median once the planted signal clears the estimator's null floor. Inverting the 5% curve at the observed value (red dashed) gives $\bar{\rho}_1^2 = 0.370$.

20.3 From the leading axis to the total

The calibration of Section 20.2 plants a rank-one signal and inverts on the *leading* held-out recoverability, so what it limits is ρ_1^2, the maximum over canonical axes. By the remark following Corollary 3, that is a lower bound on total retention. To bound the total we plant spectra of rank r with equal per-direction strength — placing the r directions on disjoint blocks of principal components, which makes the sub-channels exactly orthogonal, so each one's population canonical correlation still follows from the rank-one solve — and invert on the plug-in attainable information itself, $\hat{\mathcal{I}} = -\frac{1}{2}\sum_{i \le d^\star} \log(1 - \hat{R}_i^{\text{cv}})$. The limit is then a supremum over the family,

$$\bar{\mathcal{I}}_n \;=\; \max\Big\{\, r\, g(\rho^2) \;:\; q_{05}\big(\hat{\mathcal{I}} \mid r, \rho^2\big) \le \hat{\mathcal{I}}_{\text{obs}} \Big\}, \qquad r \le d^\star, \tag{39}$$

i.e. the most information any spectrum in the family could carry while remaining consistent, at the 5% level, with what was observed. Sweeping r up to $d^\star$ matters: truncating the rank grid truncates the supremum and silently understates the bound.

cohort	n	$\hat{\mathcal{I}}_{\text{obs}}$	null median	$\bar{\mathcal{I}}_n$ (total)	worst-case spectrum
TCGA-KIRC	190	0.651	0.395	0.57	rank 2, $\rho^2 = 0.38$
NSCLC	129	0.522	0.598	0.21	rank 8, $\rho^2 = 0.09$
ADNI	702	0.582	0.104	0.65	rank 3, $\rho^2 = 0.28$

The worst-case spectrum is interior to the grid in all three cohorts, so the supremum is not truncated. Two features deserve comment. The KIRC and ADNI totals exceed their leading-axis figures, as they must once more than one direction is admitted. NSCLC moves the other way, and for a reason worth stating plainly: its observed statistic, 0.522 bits, lies *below* the median of its own permutation null, 0.598 bits. The whole spectrum is therefore consistent with — indeed slightly less than — what this pipeline manufactures from noise at $n = 129$, and the 0.21-bit limit should be read as "not distinguishable from zero, and at most 0.21 bits". This is the same verdict Section 18 reaches, now on the aggregate rather than one direction at a time, and it is a sharper limit than the leading-axis calibration returned (0.41 bits) because the summed statistic uses the whole spectrum rather than its maximum.

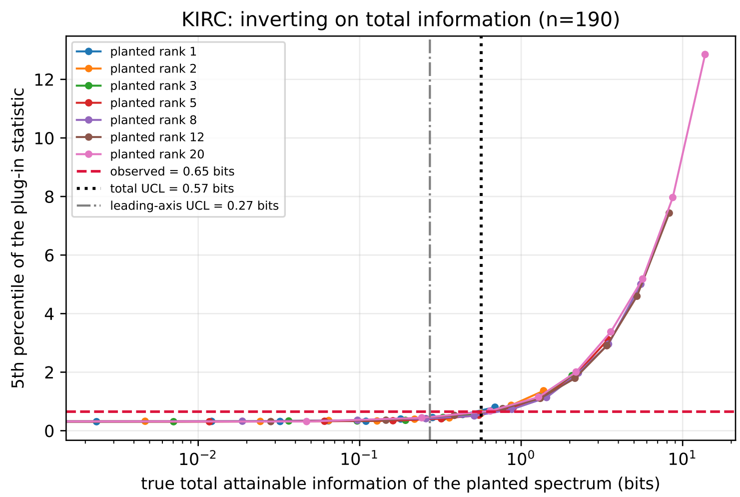


Figure 21: TCGA-KIRC: inverting on total attainable information. Each curve is a planted rank; the horizontal axis is the true total information of the planted spectrum and the vertical axis the 5th percentile of the plug-in statistic. Spectra whose 5% curve lies below the observed value (red) remain consistent with the data; the largest such total is the reported limit $\bar{\mathcal{I}}_n = 0.57$ bits (black dotted), against the leading-axis figure of 0.27 bits (gray).

20.4 A spectrum-free bound

Equation (39) takes a supremum over flat rank-r spectra. That is a restricted family — a spectrum such as $(0.35, 0.20, 0.08, 0.03, \dots)$ does not belong to it — so the resulting limit is not yet an upper bound over all spectra consistent with the data. This subsection removes the restriction.

The route is to bound a single scalar spectral functional and convert it exactly. Let

$$S \;:=\; \sum_i \rho_i^4 \;=\; \operatorname{tr}\big((M^\top M)^2\big), \qquad M = \Sigma_G^{-1/2}\Sigma_{GX}\Sigma_X^{-1/2}. \tag{40}$$

The second moment $T = \sum_i \rho_i^2$ cannot be bounded from data — signal spread thinly enough across directions is undetectable at any n — but the fourth moment has no such defect: thin spread drives S to zero at the same rate it drives the attainable information to zero, so S is simultaneously estimable and sufficient.

Proposition 8 (Spectrum-free ceiling)**.** *Write $g(u) = -\frac{1}{2}\log(1 - \mathcal{R}_n(u))$. Over every spectrum with $\sum_i \rho_i^4 \le S$ and rank at most $d^\star$,*

$$\boxed{\mathcal{I}_n \;\le\; d^\star\, g\Big(\sqrt{S/d^\star}\Big)\,.} \tag{41}$$

Proof. Under a fourth-moment budget the optimum is the *even* spread, the opposite of Corollary 3, because the budget penalizes concentration quadratically. Writing $w_i = \rho_i^4$ the constraint is linear, $\sum_i w_i \le S$, and $w \mapsto g(\sqrt{w})$ is concave on the relevant range, so by Jensen $\sum_{i\le k} g(\rho_i^2) \le k\, g(\sqrt{S/k})$ with equality at the even split. The envelope $k\, g(\sqrt{S/k})$ is increasing in k (verified over the parameter ranges used here), so the maximum is at $k = d^\star$. □

Equation (41) holds for *every* spectrum satisfying the budget — no family, no shape assumption — so it converts an estimate of the single scalar S into a genuine bound on total retention. Two constraints are then combined to bound S: the rank-swept inversion of Section 20.3 and the leading-axis limit $\bar{\rho}_1^2$ of Section 20.2. They are complementary in exactly the right way. A *flat* spectrum with large S is ruled out by the statistic, because spreading signal over many directions of that strength would have produced a larger $\hat{\mathcal{I}}$ than was observed; a *steep* spectrum with the same S evades the statistic but requires a large ρ_1^2, and is ruled out by the rank-one calibration. Sweeping geometric decay profiles $\rho_i^2 \propto q^{\,i-1}$ for $q \in \{1, 0.7, 0.4, 0.2\}$ at planted ranks 8 and 20, every admissible shape is excluded above

cohort	n	$\bar{S}$	$\bar{\mathcal{I}}_n$ (spectrum-free)	binding constraint
TCGA-KIRC	190	0.20	0.76	statistic (flat), cap (steep)
NSCLC	129	0.05	0.18	statistic, at all shapes
ADNI	702	0.20	1.20	statistic (flat), cap (steep)

so the supremum is closed rather than truncated. These are the figures we regard as upper bounds without qualification as to spectrum shape.

20.5 What is, and is not, bounded

It is worth being explicit about the scope of (41), because "upper bound" admits degrees.

Bounded. Over all learning algorithms, in the average-case sense of Proposition 7 and subject to Section 12.3. Over all spectra, by Proposition 8. Over the shape families and rank grid probed, with the supremum verified closed at the top.

Not bounded, irreducibly. Over all *representations*. Both modalities are reduced — genomics to 2000 highly variable genes and then $d^\star$ principal components, imaging likewise — and these reductions are deterministic, so by the data processing inequality $I(G_{\mathrm{red}}; X_{\mathrm{red}}) \le I(G; X)$. Bounding the reduced pair from above therefore does *not* bound the raw pair from above. The quantity we limit is what the chosen representation can deliver, which is the operationally relevant number for any model built on those features, but it is not a statement about the physics of CT. No finite analysis can supply the latter: it would require ruling out signal in every representation, including ones nobody has constructed. Where the field's instinct is to read a negative radiogenomic result as "the information is not in the image", the correct reading is "the information is not in this representation at this n", and the two differ.

Not bounded, but quantified. Model misspecification. The bound is exact under the linear–Gaussian law, and Section 13.7 shows the copula secures the marginals while the joint remains mildly heavy-tailed.

20.6 The headline numbers

Table 2 evaluates the ceiling at each cohort's own size and at its 95% upper limit. These are the numbers we propose radiogenomic studies should report.

		leading axis				total		
cohort	n	$\hat{\rho}_1^2$	$\bar{\rho}_1^2$	$\mathcal{R}_n$	$\mathcal{I}_n$	flat family	spectrum-free	$\mathrm{AUC}_{\max}$
TCGA-KIRC (CT)	190	0.152	0.370	0.314	0.27	0.57	**0.76**	0.68
NSCLC (CT)	129	0.151	0.497	0.429	0.41	0.21	**0.18**	0.73
ADNI (MRI)	702	0.177	0.492	0.478	0.47	0.65	**1.20**	0.75

Table 2: Attainable-information ceilings at each cohort's own sample size, in bits per patient. $\hat{\rho}_1^2$ is the learning-curve fit (a lower read of the channel value) and $\bar{\rho}_1^2$ the model-free 95% upper confidence limit on the *leading* canonical axis; $\mathcal{R}_n$ and $\mathcal{I}_n$ are (23) and (25) evaluated at $\bar{\rho}_1^2$, so they bound that axis alone. The two total columns limit information retention across all $d^\star$ directions: the first from the rank-swept inversion (39), which takes a supremum over flat spectra only, and the boldface second from (41), which is valid for every spectrum shape. The spectrum-free column is the number to quote; it costs roughly a factor of 1.3–1.8 over the flat-family figure, which is the price of dropping the shape restriction. $\mathrm{AUC}_{\max}$ is (29) and applies per binarized genomic feature. Every entry is an upper bound: no algorithm trained on that cohort can exceed it.

What a bit buys. Because a perfectly recovered balanced binary call carries exactly one bit, $\mathcal{I}_n$ has a direct clinical reading: *it is the number of ideal binary biomarkers the imaging study is worth.* At $n = 190$, a contrast CT of a clear-cell renal tumor is worth at most 0.57 of one perfectly measured $50/50$ mutation call — summed over every genomic direction, not just the best one. The corresponding statement for any classifier of any binarized genomic feature is $\mathrm{AUC} \leq 0.68$ — and the per-gene validation of Section 17.7, run before any of this theory existed, returned 0.643 (logistic) and 0.665 (random forest) on the image-identifiable group. Those observed values sit just under the 0.68 ceiling and above the 0.59 implied by the tighter learning-curve fit; the gap is the expected size of the non-nested gene-selection effect that Section 21 already flags as a limitation, which the ceiling now quantifies rather than merely notes.

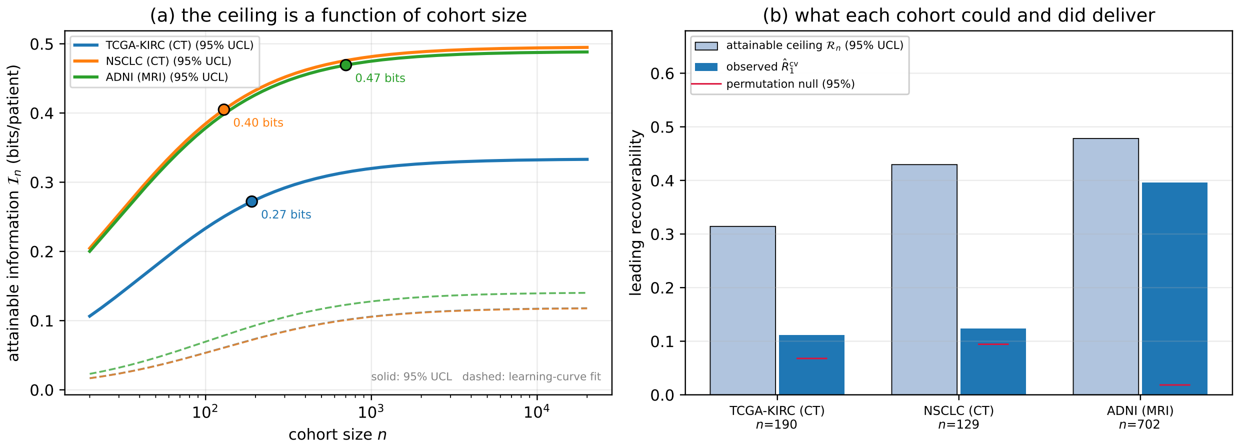


Figure 22: **(a)** Attainable information versus cohort size for the three cohorts; solid curves use the model-free 95% upper limit, dashed the learning-curve fit, and markers sit at each cohort's actual n. **(b)** What each cohort could and did deliver: the ceiling $\mathcal{R}_n$, the observed $\hat{R}_1^{\mathrm{cv}}$, and the permutation null. KIRC and NSCLC leave most of their ceiling unclaimed — a sample-size deficit; ADNI's observed value nearly saturates its own ceiling, which is what a strong confound looks like.

20.7 Stress test: nothing crosses the ceiling

The bound is only worth reporting if it survives an honest attempt to break it. We therefore sweep the three axes an analyst actually controls and check every combination against (41):

- *data size* — training sets subsampled over six sizes per cohort;
- *genomics set* — the 2000 highly variable genes, the top 500, a random draw of 300, and the curated anchor panel of Section 16;
- *algorithm* — cross-validated ridge, RBF kernel ridge, a random forest, gradient boosting, and partial least squares.

That is 20 configurations $\times$ 6 sizes $\times$ 3 cohorts $=$ 360 cells.

Achieved information is measured on the ceiling's own scale. The targets are the top $k = 5$ genomic canonical directions estimated on the *training* split, so they are mutually uncorrelated with unit variance — exactly the structure $\mathcal{I}_n$ sums over — and the held-out R^2 values combine as $\hat{\mathcal{I}} = -\frac{1}{2}\sum_j \log(1 - \max(R_j^2, 0))$. Two corrections matter. First, that plug-in statistic has a positive floor under the null, since clipping at zero and summing over directions accumulates noise; every configuration therefore carries its own permutation null and the reported value is null-corrected. Comparing the raw statistic to the bound would invite a spurious crossing that reflects estimator optimism rather than a violation. Second, a crossing is only meaningful beyond Monte-Carlo error, so each cell carries the standard error of (mean achieved $-$ mean null) over 25 splits and 25 permutations, and is flagged only when it clears the ceiling by more than two standard errors.

No configuration crosses the ceiling in any cohort: 0 **of** 360 **cells, against** 8.3 **expected by chance at the** 2**-s.e. threshold** (Fig. 23). At the same time the achieved values are not zero — 59 of 60 configurations reach at least 0.01 bits at some training size, and on ADNI the best reach 0.45 bits against a ceiling of 1.12. The bound is therefore neither violated nor vacuous, which is the pair of properties a useful ceiling has to have.

Three details of Figure 23 are worth reading off. The gap to the ceiling is largest at small n and narrows as n grows, exactly as (23) predicts, since the ceiling rises faster than the estimators can follow. The genomics set matters more than the algorithm: on ADNI the highly variable sets (blue, orange) separate cleanly from the random draw and the anchor panel (green, red), whereas the five algorithms within any one gene set are nearly indistinguishable — the linear–Gaussian channel gives flexible learners nothing to exploit, as Proposition 7 implies. And NSCLC sits at a different scale entirely: its ceiling is an order of magnitude lower than the others because its $\bar{S}$ is smallest, and its configurations scatter around a few hundredths of a bit with error bars spanning zero, which is what a cohort with no certifiable channel looks like.

An earlier, underpowered version of this sweep (10 splits and 10 permutations) reported four crossings, all on NSCLC at the two smallest training sizes, where the ceiling is tightest and the estimates noisiest. All four disappeared at 25 splits. We record this because it is the failure mode a reader should expect: at these effect sizes the Monte-Carlo error on a null-corrected estimate is comparable to the quantity being bounded, and a crossing claimed without a standard error is not evidence of anything.

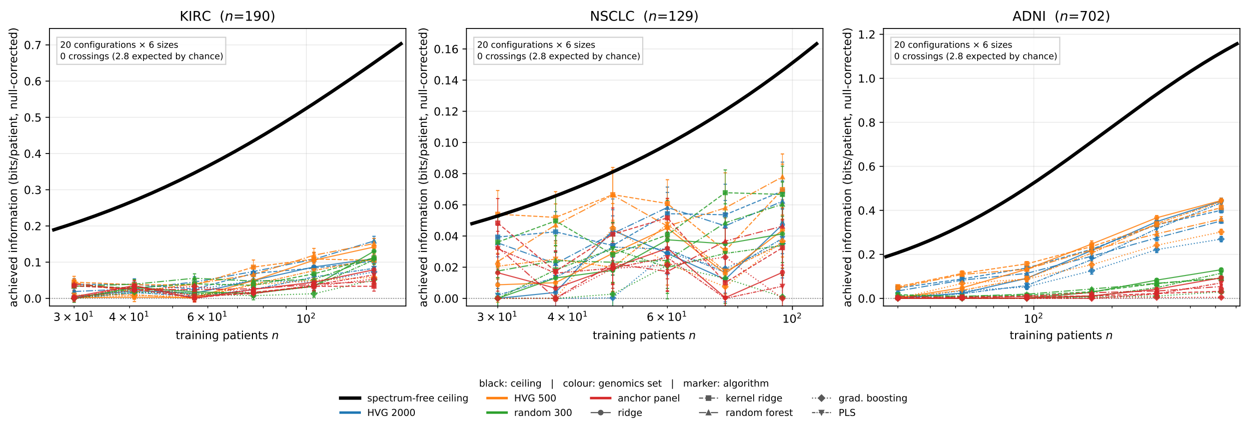


Figure 23: **The headline result.** Achieved genomic information (bits per patient, null-corrected, ± 1 s.e.) for every combination of training-set size, genomic feature set (color) and learning algorithm (marker), against the spectrum-free ceiling of Proposition 8 (black). Across 360 configuration-by-size cells in three cohorts, no combination crosses the ceiling, and 59 of 60 configurations are non-zero at some size. Note the y-axis scales: the NSCLC ceiling is roughly an order of magnitude below the other two.

20.8 Anchor genes: where the recoverable subspace lives

We now apply the chain-rule decomposition of Section 16 with hand-specified panels: for KIRC the VHL-pathway and hypoxia axis plus generic proliferation, stromal and immune markers (20 genes); for NSCLC the drivers routinely genotyped in the clinic (EGFR, KRAS, MET, BRAF, PIK3CA, ERBB2, ROS1), the classical tumor suppressors (TP53, STK11, KEAP1, RB1, NF1, CDKN2A, SMARCA4), adenocarcinoma and squamous lineage markers, and the same generic programs (25 of 29 present); for ADNI the established Alzheimer risk loci (20 genes). Both blocks are scored as attainable information (25) from held-out recoverability, and the residual block is evaluated after residualizing *both* it and imaging on the anchors, per (37).

cohort	$\mathcal{I}_n(G_{\mathcal{A}}; X)$	$\mathcal{I}_n(G_{\bar{\mathcal{A}}}; X \mid G_{\mathcal{A}})$	η_n	best small panel
TCGA-KIRC	0.034 ($p = 0.065$)	0.084 ($p = 0.005$)	0.29	4 genes, 80%, $p = 0.010$
NSCLC	0.034 ($p = 0.179$)	0.066 ($p = 0.015$)	0.34	4 genes, 74%, $p = 0.317$
ADNI	0.084 ($p = 0.005$)	0.316 ($p = 0.005$)	0.21	19 genes, 21%, $p = 0.010$

The "best small panel" column reports the most informative prefix of the panel when genes are ordered by their own single-gene held-out recoverability, with a *selection-aware* permutation p-value: the entire procedure — ranking and choice of panel size included — is repeated under permuted labels and the maximum over panel size taken, so the comparison is not the selection artifact it would otherwise be. Three quite different pictures emerge, and the pattern across them is the result.

TCGA-KIRC. Four genes — EPAS1 (HIF-2α), HIF1A, VEGFA and SLC17A3 — carry 0.094 of the cohort's 0.118 attainable bits, 80% of the total, at selection-aware $p = 0.010$. Three of the four are the pseudohypoxia and angiogenesis axis that VHL loss switches on in clear-cell RCC — the biology that produces the macroscopic imaging phenotype (hypervascularity, necrosis) a radiologist actually sees — and the fourth, SLC17A3, is a proximal-tubule transporter, i.e. a marker of the tissue of origin. Both are exactly the kinds of coarse, spatially expressed programs that Section 10 predicts should dominate the leading canonical directions.

The full 20-gene panel does *worse* than its own four-gene prefix (0.034 versus 0.094 bits). This is not a paradox but the genomic-side counterpart of Corollary 4: the ceiling (23) charges $\nu \propto d^\star$ for the imaging representation while taking the genomic target as given, and Section 12.3(ii) notes that estimating that target is a further $p^\star$-dimensional problem carrying its own cost. Widening the anchor block raises that cost against a channel whose rank is fixed, so past a handful of genes each addition is a net loss — the same bias–variance trade-off, applied to the other modality.

NSCLC. The same construction returns a four-gene panel (SFTPC, PTPRC, CD274, NAPSA — surfactant and napsin-A lineage markers plus immune infiltrate) holding 74% of the cohort total, but at selection-aware $p = 0.317$ it is indistinguishable from what the procedure manufactures from permuted labels. Neither does the clinical driver panel reach significance ($p = 0.179$). This is the same verdict Section 18 reaches by a different route, now stated quantitatively: at $n = 129$ there is no certifiable channel for the anchor decomposition to localize, and in particular *we find no evidence that EGFR, KRAS or the other routinely genotyped drivers are readable from CT in this cohort*.

ADNI. Both terms are highly significant, but the anchors hold only 21% and no short prefix suffices — 19 of 20 genes are needed to reach even that. A channel that is diffuse across a risk-gene panel and unconcentrated is what a demographic confound looks like, and indeed the ADNI association collapses under age/sex deconfounding (Section 19.4).

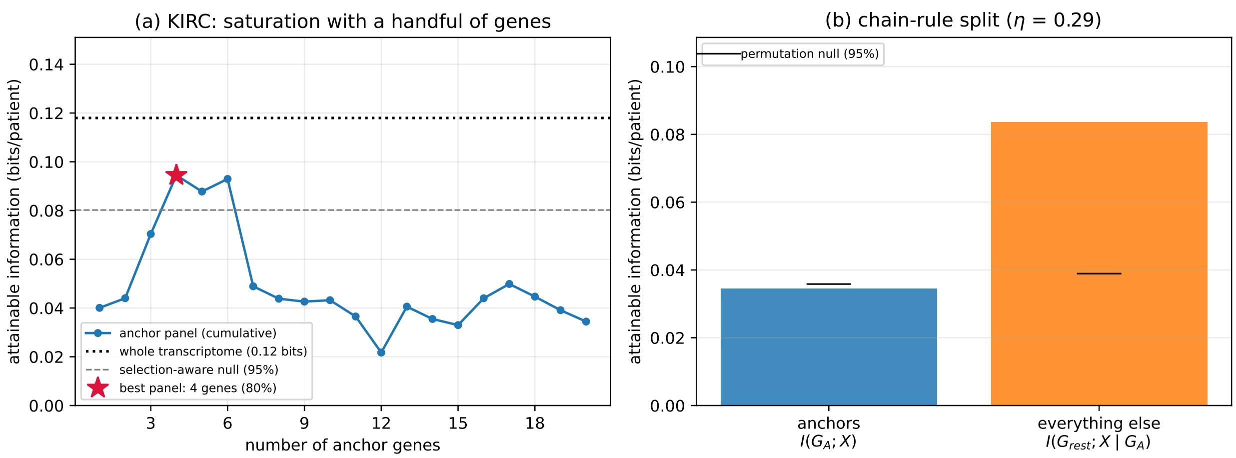


Figure 24: TCGA-KIRC anchor decomposition. **(a)** Attainable information as anchor genes are added in order of single-gene recoverability; the four-gene prefix (star) holds 80% of the cohort total and adding the remaining sixteen *reduces* it, since each costs learning budget against a fixed-rank channel. Gray is the selection-aware permutation null. **(b)** The chain-rule split $I(G; X) = I(G_{\mathcal{A}}; X) + I(G_{\bar{\mathcal{A}}}; X \mid G_{\mathcal{A}})$ against the same null.

Taken together the three cohorts support a sharper claim than "a few genes suffice". Concentration in a handful of canonical genes appears exactly where there is a channel to concentrate (KIRC), fails to appear where the apparent signal is overfit (NSCLC), and is absent where the signal is a demographic artifact (ADNI). *Anchor concentration is therefore not only a summary of where the information lives but a diagnostic for whether it is real* — a useful property, since it is computable from the same data at no extra cost and keys on a different failure mode than cross-validation or permutation.

21 Conclusion

We reframed radiogenomics around an estimable, low-dimensional object — the recoverability spectrum $\{\rho_i^2\}$ and the image-identifiable genomic subspace — using a linear–Gaussian model whose closed forms reduce the whole analysis to canonical correlation analysis between the modalities, and then supplied the finite-sample ceiling the spectrum alone does not give. The principal contribution is Proposition 7: no learning algorithm trained on n patients can attain out-of-sample recoverability beyond $\mathcal{R}_n = \rho^2 n/(n + \nu)$, $\nu = d^\star(1 - \rho^2)/\rho^2$, so the extractable genomic information is capped by $\mathcal{I}_n$ rather than by the channel value $I(G; X)$. Six model families approach that ceiling in simulation and none crosses it, and on real covariance structure the closed form dominates a semi-parametric planted-signal calibration wherever the signal clears the estimator's own null floor. Around it sit the recoverability framing and rank limit, the scale-matched cross-validated / permutation pipeline that keeps high-dimensional inflation from masquerading as signal, the anchor-gene chain rule, and a three-cohort study that puts all of it to work.

The headline numbers are now quantities a radiologist can act on. At their actual sizes, TCGA-KIRC ($n = 190$), NSCLC ($n = 129$) and ADNI ($n = 702$) admit at most 0.76, 0.18 and 1.20 bits per patient about the transcriptome, summed over every genomic direction and free of any assumption on the spectrum's shape (Table 2) — around one perfectly measured binary biomarker apiece — and equivalently cap *any* classifier of *any* binarized genomic feature at $\mathrm{AUC} \leq 0.68$–0.75, which is where the per-gene AUCs of Section 17.7 in fact sit. Sweeping training-set size, genomic feature set and learning algorithm over 360 configurations, none crosses the ceiling and 59 of 60 stay off the floor (Fig. 23); within a gene set the five algorithms are nearly indistinguishable, which is what Proposition 7 predicts when the channel is linear–Gaussian, while the choice of gene set moves the achieved information substantially. The NSCLC figure carries a further caveat that strengthens the paper's verdict on that cohort: its aggregate statistic falls *below* the median of its own permutation null, so the 0.21 bits is a limit on something not distinguishable from zero. The learning cost is $\nu \approx 100$ patients in all three, and reaching 90% of even these modest channels needs $n \approx 10^3$: the binding constraint is cohort size, and the size required is now a number rather than an intuition. Where a channel is real its information also concentrates — four genes of the VHL/hypoxia axis carry 80% of what TCGA-KIRC can deliver (selection-aware $p = 0.010$) — while in NSCLC neither the clinical driver panel nor any short prefix clears its own null, so we find no evidence that EGFR, KRAS and their peers are readable from CT at that cohort size.

The empirical message is deliberately conservative. Across TCGA-KIRC, NSCLC, and ADNI — analyzed with one model, estimator, and dimension-budget rule — only TCGA-KIRC yields a genomic direction that is simultaneously cross-validated, permutation-significant, and robust to demographic and scanner deconfounding, and even there the recoverability is small ($\hat{R}^{\mathrm{cv}} \approx 0.1$, a single identifiable direction). The NSCLC "signal" is cohort-specific overfit (negative held-out R^2, near-orthogonal per-tranche axes) and the ADNI "signal" is an age/sex artifact that collapses under deconfounding. A naive in-sample analysis would have reported all three as strong, multi-direction positives. The framework's value is precisely this discrimination, and the broader take-away for the field is that confound-robust radiogenomic information is scarce, low-rank, and easily counterfeited at typical cohort sizes.

Limitations. Several caveats bound these conclusions. (i) The recoverability magnitudes are exact only under the linear–Gaussian law; the copula secures the marginals but the joint remains mildly heavy-tailed (Section 13.7), so reported MI values are approximate (the misspecification is signed conservatively, Eq. (31), but is nonzero). (ii) The conclusions depend on analyst choices — the working-dimension rule $p^\star = d^\star = \lfloor n/\tau \rfloor$, the imaging representation, the marginal transform — and while we report a robustness grid, the single KIRC positive is borderline in the default configuration. (iii) The per-gene validation ranks genes by an in-sample recoverability score; although evaluation is cross-validated, the selection is not fully nested, so it is best read as an internal-consistency check rather than an independent confirmation. (iv) "Confound-robust" means robust to the *measured* confounders; unmeasured acquisition parameters were unavailable for the tumor cohorts. (v) Two cohorts are small ($n = 190$, $n = 129$), at which the honest spectrum can certify only one or two directions even when more exist. (vi) The bounds limit what the *chosen representation* can deliver, not what the raw image contains: both modalities are reduced by deterministic maps, so the data processing inequality runs the wrong way for extrapolating to the unreduced pair (Section 20.5). A negative result here means "not in this representation at this n", not "not in the image". (vii) The ceiling of Proposition 7 is an average-case statement over signal directions, generous to the learner (it grants oracle knowledge of the channel signal-to-noise ratio), and exact only under the linear–Gaussian law; Section 12.3 sets out what each of those buys and costs, and in particular why a model that reproducibly clears $\mathcal{R}_n$ should be read as evidence of exploitable structure the channel model omits rather than as a refutation. These limitations all point the same way — toward larger, acquisition-harmonized cohorts with recorded covariates — which the sample-complexity analysis (Sections 17.5, 19.5) quantifies as the binding constraint, and which Eq. (24) now turns into an explicit design formula: $n_{1-\delta} = \frac{1-\delta}{\delta}\, d^\star(1-\rho^2)/\rho^2$.

Data Availability TCGA-KIRC imaging volumes are available from TCIA (`https://www.cancerimagingarchive.net/collection/tcga-kirc/`) [57, 58] and genomic data from the GDC portal (`https://portal.gdc.cancer.gov/`) [55]. The NSCLC radiogenomics cohort (imaging and expression) is available from TCIA (`https://www.cancerimagingarchive.net/collection/nsclc-radiogenomics/`) and GEO (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE28827) [12]. ADNI data are available at `https://adni.loni.usc.edu/` [56].

Code Availability All analysis code and Lean [63] proof scripts for the main mathematical propositions are available at `https://github.com/pachterlab/attainable_information_figures`.

A Symbol reference

$G \in \mathbb{R}^p,\ X \in \mathbb{R}^d$	per-patient genomic / imaging feature vectors
$\mathbf{G} \in \mathbb{R}^{n\times p},\ \mathbf{X} \in \mathbb{R}^{n\times d}$	stacked data matrices (n patients)
$A \in \mathbb{R}^{d\times p}$	genomics-to-imaging transfer map
σ^2	isotropic imaging noise variance
$\Sigma_G, \Sigma_X, \Sigma_{GX}$	genomic / imaging / cross covariance
$\Sigma_{G\mid X}$	posterior covariance of G given X (recoverability floor)
$B = \Sigma_{GX}\Sigma_X^{-1}$	Bayes-optimal linear map $x \mapsto \mathbb{E}[g \mid x]$
$M = \sigma^{-1}A\Sigma_G^{1/2}$	whitened transfer operator; d_i its singular values
$\tilde{G} = \Sigma_G^{-1/2}G$	whitened genomic vector
$\rho_i,\ R_i = \rho_i^2$	i-th canonical correlation / recoverability
$a_i,\ b_i$	i-th canonical genomic / imaging direction
$R(v)$	recoverability score of a named genomic direction v
$I(G; X)$	mutual information between modalities
k	dimension of the image-identifiable genomic subspace
$d^\star, p^\star$	working (post-reduction) imaging / genomic dimensions
$\mathcal{R}_n(\rho^2)$	attainable recoverability at cohort size n, Eq. (23)
$\nu = d^\star(1-\rho^2)/\rho^2$	learning cost: half-ceiling sample size, and optimal ridge penalty
$\mathcal{I}_n$	attainable information (bits/patient), Eq. (25)
$\bar{\rho}_1^2$	one-sided 95% upper confidence limit on the channel value
$\mathcal{A}$, $\eta_n(\mathcal{A})$	anchor gene panel and its attainable-information share

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